ETSI TS 104 154 V1.1.1 (2026-08)



TECHNICAL SPECIFICATION

Core Network and Interoperability Testing (INT); Framework for a new 5G end-to-end QoS & QoE interoperability testing

Reference
DTS/INT-00206

Keywords
5G, interoperability, QoE, QoS

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from the
ETSI Search & Browse Standards application.

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format on ETSI deliver repository.

Users should be aware that the present document may be revised or have its status changed, this information is available in the Milestones listing.

If you find errors in the present document, please send your comments to the relevant service listed under Committee Support Staff.

If you find a security vulnerability in the present document, please report it through our Coordinated Vulnerability Disclosure (CVD) program.

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part of this document may be reproduced in any form, by any means and in any media, without the prior written authorization of ETSI and except as expressly permitted below.
By way of exception and when the document is a normative deliverable (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)), ETSI authorizes to reproduce and incorporate into products, services and technical documentation only those extracts (e.g. templates) that are strictly necessary for the technical implementation of the normative deliverable, to ensure compliance with the latter.
Nothing in this notice shall be construed as limiting any mandatory exceptions to copyright provided by applicable law.

© ETSI 2026.
All rights reserved.

Contents

Intellectual Property Rights 6
Foreword 6
Modal verbs terminology 6
Introduction 6
1 Scope 8
2 References 9
2.1 Normative references 9
2.2 Informative references 11
3 Definition of terms, symbols and abbreviations 11
3.1 Terms 11
3.2 Symbols 11
3.3 Abbreviations 11
4 Conventions 14
4.1 Interoperability test process 14
4.1.1 Principles 14
4.1.2 Interoperable functions statement 14
4.2 Tooling 15
4.2.1 General 15
4.2.2 The test description proforma 15
4.3 Test Description naming convention 15
5 5G Architecture Options 16
6 Configurations 17
6.1 VoNR Configurations 17
6.1.1 VoNR within one home trusted network 17
6.1.1.1 Configuration CFG_VoNR_INT_1HNW (1 home trusted network) 17
6.1.1.2 Network components 18
6.1.1.3 Protocols and Reference Points 18
6.1.2 VoNR and handover within one home trusted network 18
6.1.2.1 Configuration CFG_VoNR_INT_HO_1HNW (1 home trusted network, handover N2 or Xn) 18
6.1.2.2 Network components 19
6.1.2.3 Protocols and Reference Points 20
6.1.3 VoNR/VoWLAN within one home trusted network 20
6.1.3.1 Configuration CFG_VoNR_INT_WLAN_1HNW (1 home trusted network) 20
6.1.3.2 Network components 21
6.1.3.3 Protocols and Reference Points 21
6.1.4 VoNR/VoWLAN within two home trusted network 21
6.1.4.1 Configuration CFG_VoNR_INT_2HNW (2 home trusted networks) 21
6.1.4.2 Network components 22
6.1.4.3 Protocols and Reference Points 23
6.1.5 VoNR/VoWLAN within home and visited network case A (Roaming) 23
6.1.5.1 Configuration CFG_VoNR_RMI_2NW_A 23
6.1.5.2 Network components 24
6.1.5.3 Protocols and Reference Points 24
6.1.6 VoNR/VoWLAN within home and visited network case B (Roaming) 25
6.1.6.1 Configuration CFG_VoNR_RMI_2NW_B 25
6.1.6.2 Network components 25
6.1.6.3 Protocols and Reference Points 26
6.2 IoT Configurations 26
6.2.1 IoT 5G SA 26
6.2.1.1 Configuration CFG_5GSA 26
6.2.1.2 Network components 27
6.2.1.3 Protocols and Reference Points 27

6.2.2 IoT 5G SA Private to Public 28
6.2.2.1 Configuration CFG_5GSA_SNPN_PLMN 28
6.2.2.2 Network components 28
6.2.2.3 Protocols and Reference Points 29
6.2.3 IoT 5G SA Public to Private 29
6.2.3.1 Configuration CFG_5GSA_PLMN_SNPN 29
6.2.3.2 Network components 30
6.2.3.3 Protocols and Reference Points 30

7 Interoperable Functions Statement (IFS) 30
7.1 VoNR UE Features 30
7.2 VoNR 5GC Features 31
7.3 VoNR IMS Features 31
7.4 IoT UE Features 31
7.5 IoT 5GC features 32

8 Signalling call flows 32
8.1 Signalling call flows for voice support 32
8.1.1 UE 5G registration for VoNR support 32
8.1.2 PDU session procedures for VoNR services 34
8.1.2.1 Scope and objective 34
8.1.2.2 PDU session establishment with default QoS flow 34
8.1.2.3 PDU session modification for VoNR dedicated QoS flow 35
8.1.3 UE IMS registration 36
8.1.4 VoNR Call Establishment between two UEs 37
8.2 Signalling call flows for IoT support 38
8.2.1 IoT UE 5G registration 38
8.2.2 PDU session procedures for IoT 40

9 QoS and QoE Tests for voice over NR 42
9.1 Introduction to the concept 42
9.2 Reference network scheme 42
9.2.1 Overview 42
9.2.2 Simulation of packet loss or jitter 43
9.2.3 Simulation of codec tandeming 44
9.3 Injection and retrieval of speech test signals 44
9.3.1 Introduction 44
9.3.2 Injection and retrieval via the acoustical interface of a 5G terminal 45
9.3.3 Injection and retrieval via an IP half channel adaptor 46
9.3.4 Requirements for speech test signals 46
9.4 Selection of QoS parameters 46
9.4.1 General 46
9.4.2 End-to-end listening quality MOS-LQOF 47
9.4.2.1 Assessment compliant with P.863 47
9.4.2.2 Assessment compliant with P.565.1 47
9.4.2.3 Assessment according to proprietary methods 47
9.4.3 Mean one-way delay of speech signals 48
9.4.4 Double Talk capability 48
9.4.5 Insertion loss 48
9.4.6 Talker echo 48
9.5 Statistical analysis and reporting 49
9.5.1 General 49
9.5.2 Raw data 49
9.5.3 Data cleansing, outlier removal and expert listening 49
9.5.4 Reporting and documentation 49
9.5.5 Presentation of selected results 49
9.5.6 Specific recordings of oddities for presentation purposes 50
9.6 Classification of results 50
9.6.1 Introduction 50
9.6.2 MOS-LQOF 50
9.6.3 Mean one-way delay MOWD 51
9.6.4 Double-talk 52
9.6.5 Insertion loss 52

9.6.6 Talker Echo 52
10 QoS for IoT applications 52
10.1 Introduction 52
10.1.1 Objective 52
10.1.2 System architecture and QoS parameters (ETSI TS 123 501) 52
10.2 Relevant 5G QoS elements for IoT 52
10.3 Mapping IoT Use Cases to 5QI and Key KPIs 53
10.4 Essential end-to-end QoS test scenarios 53
10.4.1 Test Case 1 - latency validation per 5QI 53
10.4.2 Test Case 2 - Reliability & Packet Error Rate (PER) 59
10.4.3 Test Case 3 - Congestion & Priority handling 59
10.4.4 Test Case 4 - Throughput & AMBR/GBR enforcement 59
10.5 Material to be prepared for test events 59
10.6 Application specific QoS adaptors 60
10.7 Reporting 60
11 Test Descriptions 61
11.1 VoNR Test Descriptions with QoS check 61
11.1.1 Call establishment over gNB 61
11.1.2 Handover after call establishment 62
11.1.3 UE switched from gNB to WLAN after call establishment 63
11.2 IoT Test Descriptions 64
11.2.1 MQTT communication over 5GC 64
11.2.2 CoAP communication over 5GC 65
11.2.3 HTTP/REST communication over 5GC 66
Annex A (informative): Template for co-ordination of QoS options 67
Annex B (informative): Advanced environmental test conditions 68
Annex C (informative): Bibliography 69
C.1 General QoS/QoE aspects 69
C.2 Objective prediction of listening quality 69
C.3 E-Model 69
C.4 VoLTE 69
History 70

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)) may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM®** and the GSM logo are trademarks registered and owned by the GSM Association.

BLUETOOTH® is a trademark registered and owned by Bluetooth SIG, Inc.

Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Core Network and Interoperability Testing (INT).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Introduction

The deployment of 5G Standalone (5GS) networks introduces new opportunities for delivering differentiated services through advanced Quality of Service (QoS) mechanisms and service-aware network functions. At the same time, interoperability between equipment, network functions and service platforms originating from different suppliers remains essential to ensure a consistent end-user experience.

Voice over New Radio (VoNR), Voice over WLAN (VoWLAN) integration and emerging Internet of Things (IoT) applications rely on the correct interaction of User Equipment (UE), Radio Access Networks (RAN), 5G Core (5GC), IP Multimedia Subsystem (IMS) components and application servers. While individual components may conform to the relevant 3GPP and ETSI specifications, end-to-end service performance depends on the successful interoperability of all participating elements and on the correct application of QoS policies throughout the communication path.

The present document provides a framework for planning, preparing and conducting interoperability test events and demonstrations focusing on end-to-end QoS and Quality of Experience (QoE) in 5G environments. The framework covers both public and non-public 5G networks and includes voice and IoT service scenarios representative of practical deployments.

Unlike equipment conformance specifications, the present document does not define implementation requirements for individual network elements. Instead, it specifies reference configurations, interoperability procedures, test descriptions, QoS/QoE assessment methods and reporting principles that can be used during multi-vendor interoperability events. The objective is to verify that interoperable 5G systems are capable of establishing, maintaining and delivering services with the expected service quality from an end-user perspective.

1 Scope

The present document specifies a framework for interoperability testing of 5G Standalone (5GS) networks with a focus on end-to-end Quality of Service (QoS) and Quality of Experience (QoE).

The present document defines:

- reference architectures and interoperability configurations for 5G voice and IoT services;
- interoperability testing procedures applicable to multi-vendor environments;
- Interoperable Functions Statements (IFS);
- signalling procedures and reference call flows relevant to the tested scenarios;
- methods for the assessment of end-to-end QoS and QoE characteristics;
- test descriptions suitable for interoperability events, demonstrations and showcases;
- reporting principles for documenting interoperability and service quality results.

The scope includes Voice over New Radio (VoNR), VoWLAN integration with 5G systems, selected roaming and interconnection scenarios, and IoT communication over public and non-public 5G networks.

For voice services, the present document addresses:

a) successful registration and service establishment procedures;

b) correct negotiation and application of QoS-related parameters;

c) end-to-end service interoperability across participating network elements;

d) objective assessment of voice quality and related QoE indicators.

For IoT services, the present document addresses:

a) successful registration and connectivity procedures;

b) correct application of 5G QoS mechanisms;

c) end-to-end interoperability between IoT devices, networks and applications;

d) assessment of QoS-related performance indicators relevant to the supported application.

The present document does not specify conformance requirements for individual network elements, interfaces, protocols or products. Conformance testing of equipment against ETSI, 3GPP or other standards is outside the scope of the present document.

The present document also does not specify deployment requirements, commercial service requirements or minimum performance levels for operational 5G networks.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] Recommendation ITU-T G.107 (06/2015): "The E-model: a computational model for use in transmission planning".

[2] Recommendation ITU-T G.107.1 (06/2019): "Wideband E-model".

[3] Recommendation ITU-T G.107.2 (03/2023): "Fullband E-model".

[4] Recommendation ITU-T G.113 (09/2024): "Transmission impairments due to speech processing".

[5] Recommendation ITU-T G.114 (05/2003): "One-way transmission time".

[6] Recommendation ITU-T G.131 (11/2003): "Talker echo and its control".

[7] Recommendation ITU-T G.711 (11/1988): "Pulse code modulation (PCM) of voice frequencies".

[8] Recommendation ITU-T P.310 (06/2009): "Transmission characteristics for narrow-band digital handset and headset telephones".

[9] Recommendation ITU-T P.501 (04/2025): "Test signals for use in telephony and other speech-based applications".

[10] Recommendation ITU-T P.565.1 (11/2021): "Machine learning model for the assessment of transmission network impact on speech quality for mobile packet-switched voice services".

[11] Recommendation ITU-T P.800 (08/1996): "Methods for subjective determination of transmission quality".

[12] Recommendation ITU-T P.863 (03/2018): "Perceptual objective listening quality prediction".

[13] Recommendation ITU-T P.863.1 (06/2019): "Application guide for Recommendation ITU-T P.863".

[14] Recommendation ITU-T P.1010 (07/2004): "Fundamental voice transmission objectives for VoIP terminals and gateways".

[15] Recommendation ITU-T P.1401 (01/2020): "Methods, metrics and procedures for statistical evaluation, qualification and comparison of objective quality prediction models".

[16] ETSI TS 102 425 (V1.1.1): "Speech and multimedia Transmission Quality (STQ); Digital reference point for speech communication in packet based networks".

[17] ETSI TS 102 250-6 (V1.3.1): "Speech and multimedia Transmission Quality (STQ); QoS aspects for popular services in mobile networks; Part 6: Post processing and statistical methods".

[18] ETSI TS 103 739 (V1.4.1): "Speech and multimedia Transmission Quality (STQ); Transmission requirements for wideband mobile wireless terminals (handset and headset) from a QoS perspective as perceived by the user".

[19] ETSI TS 103 905 (V1.1.1): "Core Network and Interoperability Testing (INT); VoNR/ViNR and VoLTE/ViLTE interconnection testing for interworking and roaming scenarios with QoS/QoE".

[20] ETSI TS 122 186 (V19.0.0): "5G; Service requirements for enhanced V2X scenarios (3GPP TS 22.186 version 19.0.0 Release 19)".

[21] ETSI TS 122 261 (V19.12.0): "5G; Service requirements for the 5G system (3GPP TS 22.261 version 19.12.0 Release 19)".

[22] ETSI TS 123 060 (V19.0.0): "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); General Packet Radio Service (GPRS); Service description; Stage 2 (3GPP TS 23.060 version 19.0.0 Release 19)".

[23] ETSI TS 123 228 (V19.5.0): "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; IP Multimedia Subsystem (IMS); Stage 2 (3GPP TS 23.228 version 19.5.0 Release 19)".

[24] ETSI TS 123 285 (V19.0.0): "Universal Mobile Telecommunications System (UMTS); LTE; Architecture enhancements for V2X services (3GPP TS 23.285 version 19.0.0 Release 19)".

[25] ETSI TS 123 287 (V19.0.0): "5G; Architecture enhancements for 5G System (5GS) to support Vehicle-to-Everything (V2X) services (3GPP TS 23.287 version 19.0.0 Release 19)".

[26] ETSI TS 123 501 (V18.11.0): "5G; System architecture for the 5G System (5GS) (3GPP TS 23.501 version 18.11.0 Release 18)".

[27] ETSI TS 123 502 (V16.19.0): "5G; Procedures for the 5G System (5GS) (3GPP TS 23.502 version 16.19.0 Release 16)".

[28] ETSI TS 123 503 (V18.10.0): "5G; Policy and charging control framework for the 5G System (5GS); Stage 2 (3GPP TS 23.503 version 18.10.0 Release 18)".

[29] ETSI TS 124 229 (V19.4.1): "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3 (3GPP TS 24.229 version 19.4.1 Release 19)".

[30] ETSI TS 124 501 (V18.12.0): "5G; Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3 (3GPP TS 24.501 version 18.12.0 Release 18)".

[31] ETSI TS 126 238 (V19.0.0): "Universal Mobile Telecommunications System (UMTS); LTE; 5G; Uplink streaming (3GPP TS 26.238 version 19.0.0 Release 19)".

[32] ETSI TS 138 300 (V19.0.0) : "5G; NR; NR and NG-RAN Overall description; Stage-2 (3GPP TS 38.300 version 19.0.0 Release 19)".

[33] ETSI TS 138 306 (V19.0.0): "5G; NR; User Equipment (UE) radio access capabilities (3GPP TS 38.306 version 19.0.0 Release 19)".

[34] GSMA™: "5G Implementation Guidelines: SA Option 2".

[35] OASIS: "MQTT Version 5.0".

[36] IETF RFC 7252: "The Constrained Application Protocol (CoAP)".

[37] IETF RFC 9110: "HTTP Semantics".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI EG 202 798 (V1.1.1): "Intelligent Transport Systems (ITS); Testing; Framework for conformance and interoperability testing".

[i.2] ETSI ETR 266 (1996-08): "Methods for Testing and Specification (MTS); Test Purpose style guide".

[i.3] ETSI TR 126 939 (V19.0.0): "Universal Mobile Telecommunications System (UMTS); LTE; 5G; Guidelines on the Framework for Live Uplink Streaming (FLUS) (3GPP TR 26.939 version 19.0.0 Release 19)".

3 Definition of terms, symbols and abbreviations

3.1 Terms

Void.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

3GPP	3rd Generation Partnership Project
4G	4th Generation
5G	5th Generation
5GC	5G Core
5GCN	5G Core Network
5G-CP	5G Control Plane
5GMM	5G Mobility Management
5GS	5G System
ACR	Absolute Category Rating
AGW	Access Gateway
AMBR	Aggregate Maximum Bit Rate
AMF	Access and Mobility Management Function
AMR	Adaptive MultiRate
AMR-WB	AMR-WideBand
ARP	Allocation and Retention Priority
AS	Autonomous System
CFG	(Test) ConFiGuration
CIoT	Cellular Internet of Things
CN	Core Network
CoAP	Constrained Application Protocol
CP	Control Plane
CQEF	Communication Quality Estimated Fullband

CSCF	Call Session Control Function
DC	Dual Connectivity
DN	Data Network
DNN	Data Network Name
DRX	Discontinuous Reception
DSP	Digital Signal Processor
E2E	End-to-End
eDRX	Extended Discontinuous Reception
EFR	Enhanced FullRate
EHC-CP	Ethernet Header Compression Convergence Protocol
eLTE	enhanced Long-Term Evolution
eMBB	enhanced Mobile BroadBand
EN-DC	E-UTRA New Radio Dual Connectivity
EPC	Evolved Packet Core
EPS	Evolved Packet System
EUT	Equipment Under Test
E-UTRA	Evolved Universal Terrestrial Radio Access
EVS	Enhanced Voice Service
FB	FullBand
GBR	Guaranteed Bit Rate
GEO	Geostationary Earth Orbit
GFBR	Guaranteed Flow Bit Rate
GPRS	General Packet Radio Service
GSM	Global System for Mobile communications
GSMA	GSM Association
GTP	GPRS Tunnelling Protocol
HSS	Home Subscriber Server
HTTP	HyperText Transfer Protocol
HTTP2	HyperText Transfer Protocol version 2
I/S	Interworking/System
IBCF	Interconnection Border Control Function
I-CSCF	Interrogating CSCF
ID	Identifier
IE	Information Element
IFS	Interoperable Functions Statement
IFS_ID	IFS IDentifier
IMS	IP Multimedia Subsystem
IO	Input/Output
IoT	Internet of Things
IP	Internet Protocol
IPHC	IP Header Compression
IWK	InterWorKing
KPI	Key Performance Indicator
LEO	Low Earth Orbit
LQE	Listening Quality Estimated
LQEF	Listening Quality Estimated Fullband
LQO	Listening Quality Objective
LQOF	Listening Quality Objective Fullband
LQSF	Listening Quality Subjective Fullband
LTE	Long Term Evolution
MBMS	Multimedia Broadcast Multicast Service
MCPTT	Mission Critical Push-to-Talk
MDBV	Maximum Data Burst Volume
mMTC	massive Machine-Type Communication
MOS	Mean Opinion Score
MOWD	Mean One Way Delay
MS	Mobile Station
MTU	Maximum Transmission Unit
N3	N3 Interface (NG-RAN to UPF)
N3GPP	Non-3GPP
NAS	Non-Access-Stratum
NB	NarrowBand

NE-DC	NR-E-UTRA Dual Connectivity
NGAP	Next Generation Application Protocol
NGEN	Next Generation Enterprise Network
NG-RAN	Next Generation Radio Access Network
NIDD	Non-IP Data Delivery
NPN	Non-Public Network
NR	New Radio
NSA	Non-Standalone
NSSAI	Network Slice Selection Assistance Information
OLR	Overall Loudness Rating
OTT	Over The Top
PCF	Policy Control Function
PCM	Pulse Code Modulation
P-CSCF	Proxy Call Session Control Function
PDB	Packet Delay Budget
PDU	Protocol Data Unit
PER	Packet Error Rate
PLMN	Public Land Mobile Network
POLQA	Perceptual Objective Listening Quality Assessment
PS	Packet Switched
QFI	QoS Flow Identifier
QoE	Quality of Experience
QoS	Quality of Service
RAN	Radio Access Network
RAT	Radio Access Technology
RLR	Receive Loudness Rating
RRC	Radio Resource Control
RSU	Road Side Unit
RTP	Real Time Protocol
SA	StandAlone
SCG	Secondary Cell Group
S-CSCF	Serving CSCF
SIP	Session Initiation Protocol
SLR	Send Loudness Rating
SMF	Session Management Function
SNPN	Standalone Non-Public Network
SSC	Session and Service Continuity
SST	Slice Service Type
SWB	Super WideBand
TCP	Transmission Control Protocol
TD	Test Description
TELR	Talker Echo Loudness Rating
UE	User Equipment
UE-NR	User Equipment New Radio
UL	UpLink
UPF	User Plane Function
V2X	Vehicle-to-Everything
VoIP	Voice over IP
VoLTE	Voice over LTE
VoNR	Voice over New Radio
VoPS	Voice over Packet Switched
VoWLAN	Voice over WLAN
WB	WideBand
WLAN	Wireless LAN

4 Conventions

4.1 Interoperability test process

4.1.1 Principles

The goal of interoperability tests is to check that devices resulting from protocol implementations are able to work together and provide the functionalities provided by the protocols. As necessary, one message may be checked during a test, for instance, when a successful functional verification may result from an incorrect behaviour. Detailed protocol checks are part of the conformance testing process and are thus avoided during the interoperability tests.

4.1.2 Interoperable functions statement

The "Interoperable Functions Statement" (IFS) identifies the standardized functions of an EUT. These functions can be mandatory, optional or conditional (depending on other functions), and depend on the role played by the EUT.

The IFS can also be used as a pro-forma by a vendor to identify the functions that its EUT will support when interoperating with corresponding functions from other vendors.

Item column

The item column contains a number, which identifies the item.

Item description column

The item description column describes in free text each respective item (e.g. parameters, timers, etc.). It implicitly means "is <item description> supported by the implementation?".

IFS ID column

The IFS ID column defines an identifier for this particular IFS item. The IFS ID is in the Test Description field "Applicability" to select/deselect the execution of a test.

Status column

The following notations are used for the status column:

m	mandatory - the capability is required to be supported.
o	optional - the capability may be supported or not.
n/a	not applicable - in the given context, it is impossible to use the capability.
x	prohibited (excluded) - there is a requirement not to use this capability in the given context.
o.i	qualified optional - for mutually exclusive or selectable options from a set. "i" is an integer which identifies a unique group of related optional items and the logic of their selection, which is defined immediately following the table.
c.i	conditional - the requirement on the capability ("m", "o", "x" or "n/a") depends on the support of other optional or conditional items. "i" is an integer identifying a unique conditional status expression which is defined immediately following the table.
i	irrelevant (out-of-scope) - capability outside the scope of the reference specification. No answer is requested from the supplier.

NOTE: This use of "i" status is not to be confused with the suffix "i" to the "o" and "c" statuses above.

Support column

The support column shall be filled in by the supplier of the implementation using the following notations:

Y or y	supported by the implementation.
N or n	not supported by the implementation.
N/A or n/a	no answer required (allowed only if the status is n/a, directly or after evaluation of a conditional status).

4.2 Tooling

4.2.1 General

Message monitoring solutions, including audio recording and event logging, where supported, may be used to facilitate the resolution of any interoperability and/or performance issues that may be encountered during interoperability testing.

4.2.2 The test description proforma

The test descriptions are provided in proforma tables following the format described in ETSI EG 202 798 [i.1] and ETSI ETR 266 [i.2]. The following different test event types are considered during the test execution:

- A **stimulus** corresponds to an event that enforces an EUT to proceed with a specific protocol action, like sending a message, for instance.
- A **verify** consists of verifying that the EUT behaves according to the expected behaviour (for instance, the EUT behaviour shows that it receives the expected message).
- A **configure** corresponds to an action to modify the EUT configuration.
- A **check** ensures the receipt of protocol messages on reference points, with valid content. This "check" event type corresponds to the interoperability testing with the conformance check method.
- A **obtain result** corresponds to conducting a QoS specific test which provides a measurement result.
- A **verify advanced** corresponds to the availability of an advanced acoustic test environment.
- A **create** corresponds to the production of a QoS related test report. This takes place offline and not as part of the connection.
- A **conduct** corresponds to conducting the comparison of test results obtained with the classification anchor points provided. This takes place offline and not as part of the connection.

4.3 Test Description naming convention

Table 1: TD naming convention

TD_<root>_[gr1] _[gr2] _<nn>		
<root> = root applicability	VoNR	Voice over NR test
	IoT	IoT test
[gr1] = group 1	CALL	IMS session establishment for QoS
	5GEST	5G registration and PDU establishment for IoT QoS
<nn> = sequential number	01 to 99	Sequential numbers

5 5G Architecture Options

Figure 1 shows the architecture considered in the present document.

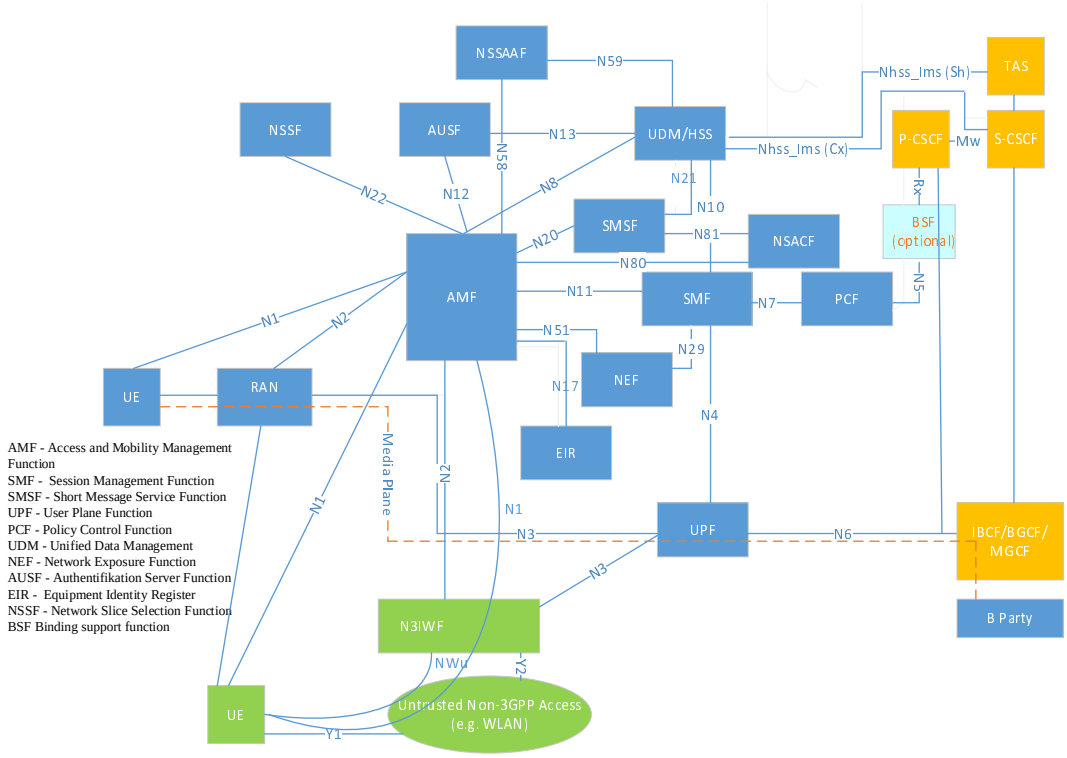


Figure 1: Non-Roaming 5G System Architecture in reference point representation

For details see clause 4.5 of ETSI TS 103 905 [19].

Table 2 shows the possible 5G SA test options, while Table 3 shows the options selected for the present document.

Table 2: 5G SA test options

Connectivity option	Core Network	Master RAT	Secondary RAT	3GPP term
Option 1 SA	EPC	LTE		LTE
Option 2 SA	**5GC**	**NR**	**-**	**NR**
Option 3 NSA	EPC	LTE	NR	EN-DC (Evolved-Universal Terrestrial Radio Access-New Radio) **NR for data only**
Option 4 NSA	5GC	NR	eLTE	NE-DC (Evolved-Universal Terrestrial Radio Access-New Radio)
Option 5 SA	5GC	eLTE	-	eLTE enhanced LTE (*eLTE*) eNB which can communicate with *5G* Core (5GC) and gNB
Option 7 NSA	5GC	eLTE	NR	NGEN-DC **NR for data only**

Table 3: Testing scenarios

	Manufacturer X Option 1 SA	Manufacturer X Option 2 SA	Manufacturer X Option 2 SA VoWLAN	Manufacturer X Option 2 SA for private network	Manufacturer X IoT
Manufacturer Y Option 1 SA	--				
Manufacturer Y Option 2 SA	Interconnection and Roaming Tests; (optional)	Interconnection and Roaming Tests;	VoWLAN Tests	Interconnection and Roaming Tests; (optional)	--
Manufacturer Y Option 2 SA VoWLAN	VoWLAN Tests (optional)	VoWLAN Tests	VoWLAN Tests	--	--
Manufacturer Y Option 2 SA for private network	Interconnection and Roaming Tests; (optional)	Interconnection and Roaming Tests;		Interconnection and Roaming Tests;	--
Manufacturer Y IoT	--	--	--	--	

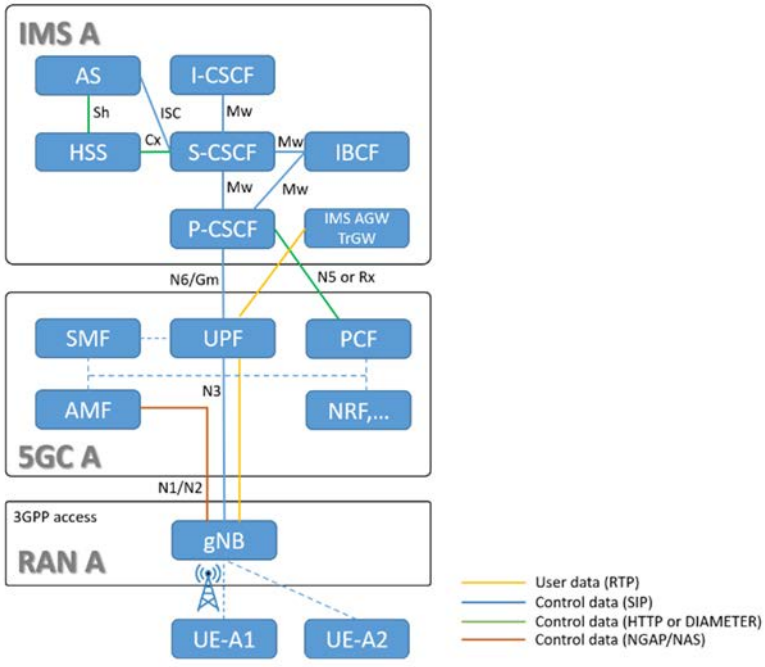
6 Configurations

6.1 VoNR Configurations

6.1.1 VoNR within one home trusted network

6.1.1.1 Configuration CFG_VoNR_INT_1HNW (1 home trusted network)

The architecture for the basic interoperability test case is shown in Figure 2.



NOTE: The Gm interface (UE to P-CSCF) is a focus for testing. In Figure 2 the Gm interface is represented as N6/Gm since it is going via the UPF N6 interface.

Figure 2: Configuration CFG_VoNR_INT_1HNW

Configuration CFG_VoNR_INT_1HNW is used for one peer network where users are 5G and IMS registered to their home network. The suffix INT stands for a home interoperability scenario. UE-A1 and UE-A2 connect to the home network A represented by 5GC A and IMS A. 5G Registration, IMS Registration, 5G Deregistration, IMS Deregistration, call establishment, call modification and call release signalling procedures of each user are performed locally in their own home network.

6.1.1.2 Network components

The following components are required for this scenario:

- Two 5G UE with IMS User Agents
- One gNB
- One 5GC with at least all of the following:
  - One AMF
  - One SMF
  - One UPF
  - One PCF
- One IMS with at least all of the following:
  - One P-CSCF
  - One I-CSCF/S-CSCF
  - One HSS
  - One AS (e.g. MMTel Application Server)
  - One IMS AGW/TrGW

6.1.1.3 Protocols and Reference Points

The following references will be interoperated in this scenario.

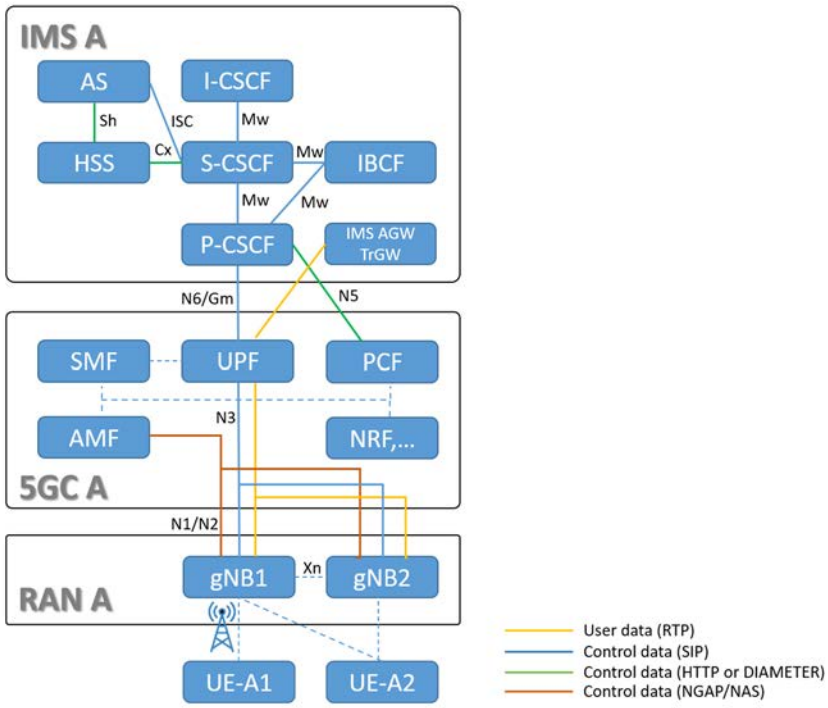
Interface	Protocol
N1/N2	5G NAS/NGAP
N5	HTTP2
Rx	Diameter
Gm	IMS SIP

NOTE: One of N5 or Rx should be supported for this configuration.

6.1.2 VoNR and handover within one home trusted network

6.1.2.1 Configuration CFG_VoNR_INT_HO_1HNW (1 home trusted network, handover N2 or Xn)

The architecture for the Basic Interoperability within one home trusted network and related to handover test case is shown in Figure 3.



NOTE: The Gm interface (UE to P-CSCF) is a focus for testing. In Figure 3 the Gm interface is represented as N6/Gm since it is going via the UPF N6 interface.

Figure 3: Configuration CFG_VoNR_INT_HO_1HNW

Configuration CFG_VoNR_INT_HO_1HNW is used for one peer network where users are 5G and IMS registered to their home network. The suffix INT stands for a home interoperability scenario. UE-A1 and UE-A2 are connected to home network A represented by 5GC A and IMS A. 5G Registration, IMS Registration, 5G Deregistration, IMS Deregistration, call establishment, call modification, call release, N2 handover and Xn handover signalling procedures of each user are performed locally in their own home network.

6.1.2.2 Network components

The following components are required for this scenario:

- Two 5G UE with IMS User Agents
- Two gNB's
- One 5GC with at least all of the following:
  - One AMF
  - One SMF
  - One UPF
  - One PCF
- One IMS with at least all of the following:
  - One P-CSCF
  - One I-CSCF/S-CSCF
  - One HSS
  - One AS (e.g. MMTel Application Server)
- One IMS AGW/TrGW

6.1.2.3 Protocols and Reference Points

The following references will be interoperated in this scenario.

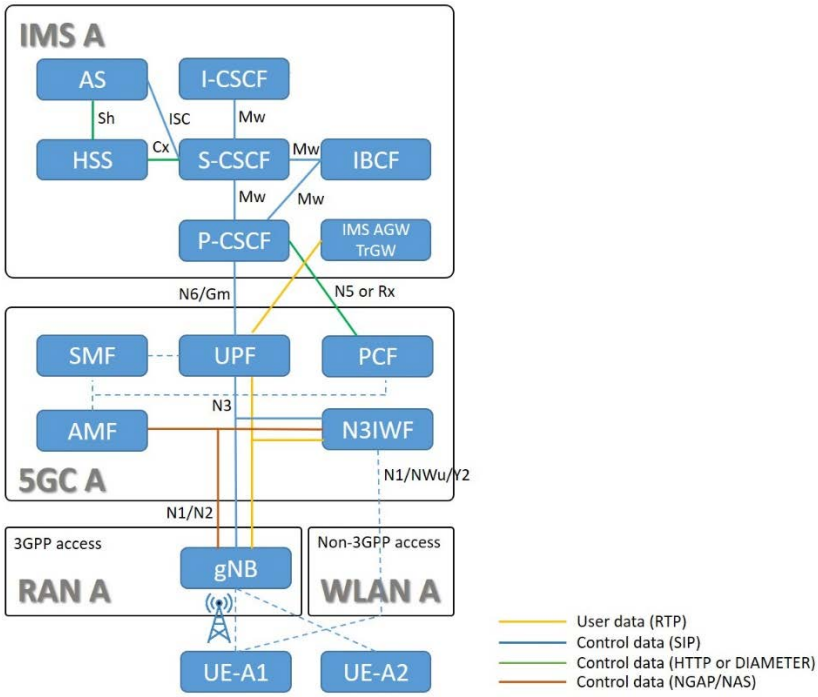
Interface	Protocol
N1/N2	5G NAS/NGAP
N5	HTTP2
Rx	Diameter
Gm	IMS SIP

NOTE: One of N5 or Rx should be supported for this configuration.

6.1.3 VoNR/VoWLAN within one home trusted network

6.1.3.1 Configuration CFG_VoNR_INT_WLAN_1HNW (1 home trusted network)

The architecture for the Basic Interoperability within one home trusted network and related to handover test case is shown in Figure 4.



NOTE: The Gm interface (UE to P-CSCF) is a focus for testing. In Figure 4 the Gm interface is represented as N6/Gm since it is going via the UPF N6 interface.

Figure 4: Configuration CFG_VoNR_INT_WLAN_1HNW

Configuration CFG_VoNR_INT_WLAN_1HNW is used for one peer network where users are 5G and IMS registered to their home network. The suffix INT stands for home interoperability scenario. UE-A1 and UE-A2 are connected to home network A represented by 5GC A and IMS A. 5G Registration, IMS Registration, 5G Deregistration, IMS Deregistration, call establishment, call modification, call release signalling procedures of each user are performed locally in their own home network. Network connectivity is changed from RAN to WLAN.

6.1.3.2 Network components

The following components are required for this scenario:

- Two 5G UE with IMS User Agents
- One gNB
- One 5GC with at least all of the following:
  - One AMF
  - One SMF
  - One UPF
  - One PCF
  - One N3IWF
- One WLAN
- One IMS with at least all of the following:
  - One P-CSCF
  - One I-CSCF/S-CSCF
  - One HSS
  - One AS (e.g. MMTel Application Server)
- One IMS AGW/TrGW

6.1.3.3 Protocols and Reference Points

The following references will be interoperated in this scenario.

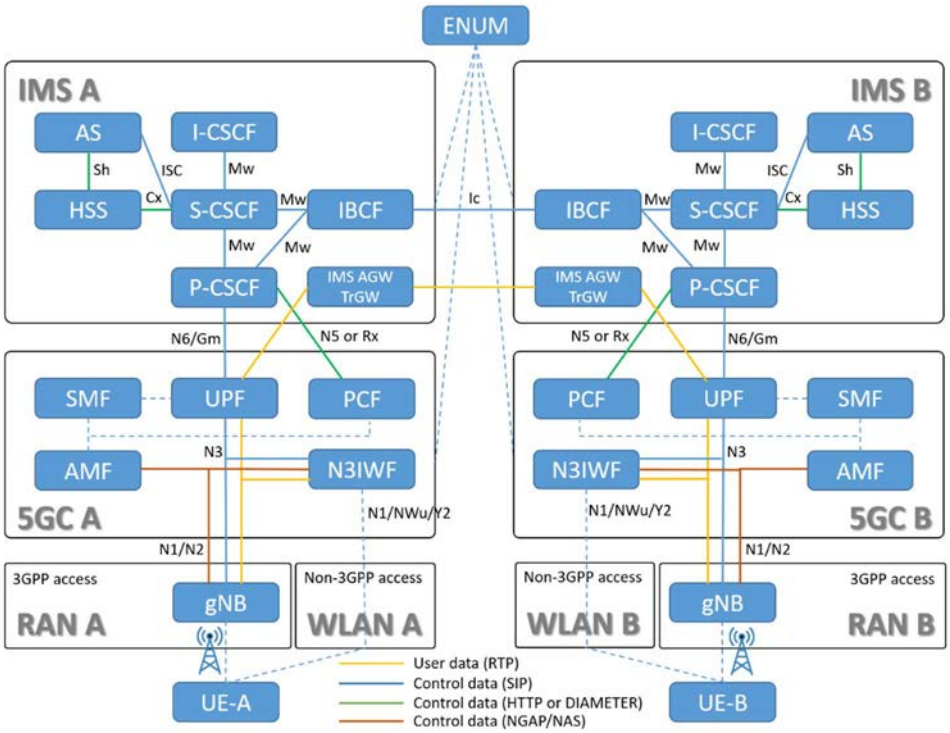
Interface	Protocol
N1/N2	5G NAS/NGAP
N5	HTTP2
Rx	Diameter
Gm	IMS SIP

NOTE: One of N5 or Rx should be supported for this configuration.

6.1.4 VoNR/VoWLAN within two home trusted network

6.1.4.1 Configuration CFG_VoNR_INT_2HNW (2 home trusted networks)

The architecture for the Interconnect test case is where two subscribers, of different operators, perform an end to end call whilst in their respective PLMN's. It is shown in Figure 5.



NOTE: The Gm interface (UE to P-CSCF) is a focus for testing. In the Figure 5 the Gm interface is represented as N6/Gm since it is going via the UPF N6 interface.

Figure 5: Configuration CFG_VoNR_INT_2HNW

Configuration CFG_VoNR_INT_2HNW is used for two peer networks where users are 5G and IMS registered to their home network. The suffix INT stands for a home interoperability scenario. UE-A connects to home network A represented by 5GC A and IMS A. UE-B connects to home network B represented by 5GC B and IMS B. 5G Registration, IMS Registration, 5G Deregistration, IMS Deregistration signalling procedures of each user are performed locally in their own home network. Call establishment, call modification and call release signalling procedures are going between the two networks over the Ici interface and therefore all related TDs are named as home interoperability tests.

6.1.4.2 Network components

The following components are required for this scenario:

- Two 5G UE with IMS User Agents
- Two gNB
- Two 5GC with at least all of the following:
  - One AMF
  - One SMF
  - One UPF
  - One PCF
  - One N3IWF
- Two WLAN

- Two IMS with at least all of the following:
  - One P-CSCF
  - One I-CSCF/S-CSCF
  - One HSS
  - One AS (e.g. MMTel Application Server)
  - One IMS AGW/TrGW
  - One IBCF

6.1.4.3 Protocols and Reference Points

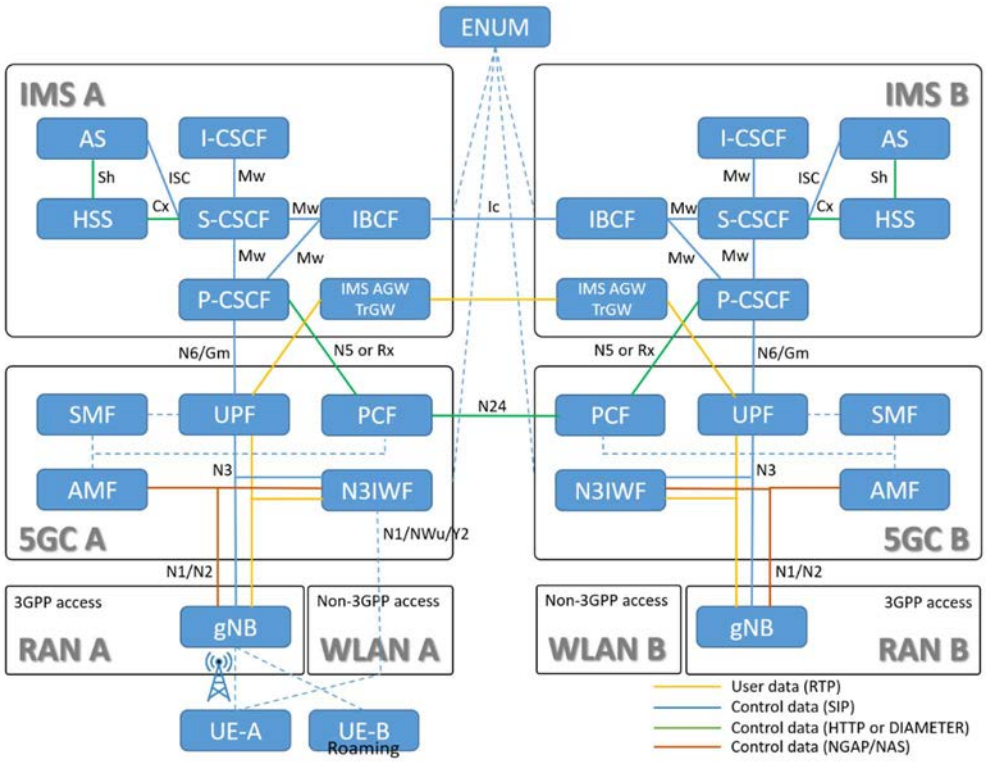
The following references will be interoperated in this scenario.

Interface	Protocol
N1/N2	5G NAS/NGAP
N5	HTTP2
Rx	Diameter
Gm	IMS SIP
Ic	IMS SIP

NOTE: One of N5 or Rx should be supported for this configuration.

6.1.5 VoNR/VoWLAN within home and visited network case A (Roaming)

6.1.5.1 Configuration CFG_VoNR_RMI_2NW_A



NOTE: The Gm interface (UE to P-CSCF) is a focus for testing. In Figure 6 the Gm interface is represented as N6/Gm since it is going via UPF N6 interface.

Figure 6: Configuration CFG_VoNR_RMI_2NW_A

Configuration CFG_VoNR_RMI_2NW_A describes the first roaming scenario. Within CFG_VoNR_RMI_2NW_A, UE-A connects to its home network A represented by 5GC A and IMS A. UE-B connects to the visited network A 5G registered to the 5GC A. 5G Registration, IMS Registration, 5G Deregistration, IMS Deregistration signalling procedures of user UE-A are performed in its own home network. 5G Registration and 5G Deregistration of UE-B is performed at the visited network A and provides the ability to subsequently register (IMS) the visiting user UE-B at the home network. UE-A acts as originating user and when a call is established towards user B the signalling runs from UE-A over its home network A towards the network of UE-B. Due to the previous IMS registration of user B, network B knows that UE-B is located at network A and signalling messages are routed back to the network A and then delivered towards UE-B. The related roaming interoperability configuration is named CFG_VoNR_RMI_2NW_A; where the suffix A signifies 'visited network A'.

6.1.5.2 Network components

The following components are required for this scenario:

One set each for PLMN-A and for PLMN B of the following:

- One 5G UE with IMS User Agents
- One gNB
- One 5GC with at least all of the following:
  - One AMF
  - One SMF
  - One UPF
  - One PCF
  - One N3IWF
- One WLAN
- One IMS with at least all of the following:
  - One P-CSCF
  - One I-CSCF/S-CSCF
  - One HSS
  - One AS (e.g. MMTel Application Server)
  - One IMS AGW/TrGW
  - One IBCF

6.1.5.3 Protocols and Reference Points

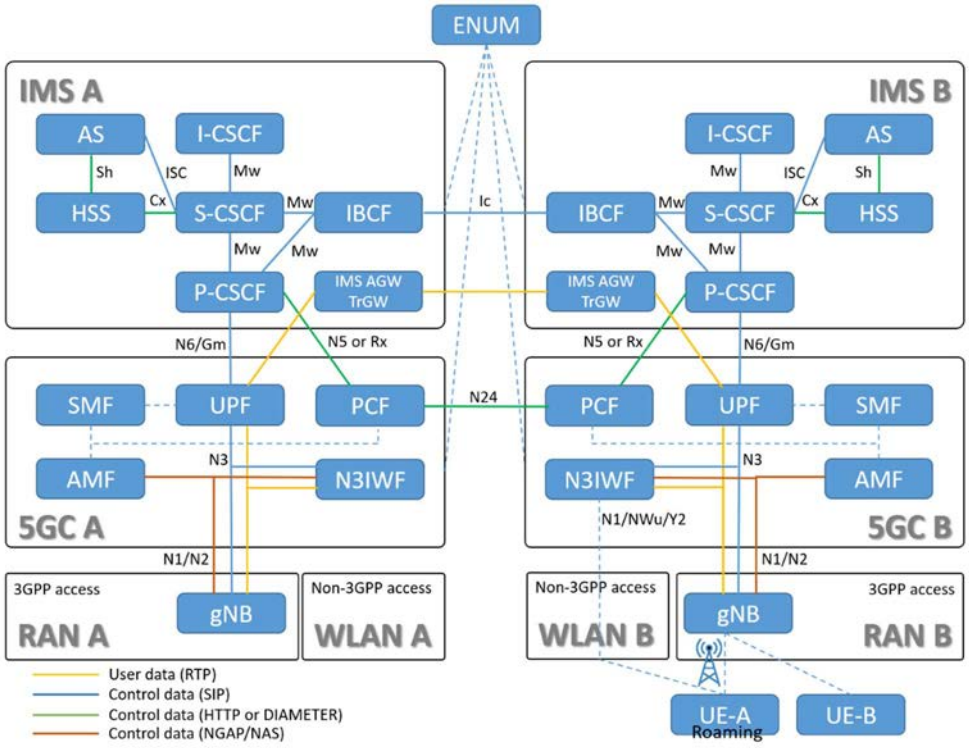
The following references will be interoperated in this scenario.

Interface	Protocol
N1/N2	5G NAS/NGAP
N5	HTTP2
Rx	Diameter
Gm	IMS SIP
Ic	IMS SIP

NOTE: One of N5 or Rx should be supported for this configuration.

6.1.6 VoNR/VoWLAN within home and visited network case B (Roaming)

6.1.6.1 Configuration CFG_VoNR_RMI_2NW_B



NOTE: The Gm interface (UE to P-CSCF) is a focus for testing. In Figure 7 the Gm interface is represented as N6/Gm since it is going via UPF N6 interface.

Figure 7: Configuration CFG_VoNR_RMI_2NW_B

Configuration CFG_VoNR_RMI_2NW_B describes the second roaming scenario. Within CFG_VoNR_RMI_2NW_B, UE-B connects to its home network B represented by 5GC B and IMS B. UE-A connects to the visited network B 5G registered to the 5GC B. 5G Registration, IMS Registration, 5G Deregistration, IMS Deregistration signalling procedures of user UE-B are performed in its own home network. 5G Registration and 5G Deregistration of UE-A is performed at the visited network B and provides the ability to subsequently register (IMS) the visiting user UE-A at the home network. UE-A acts as originating user and when a call is established towards user B the signalling runs from UE-A over visited network B towards the network of UE-A. Afterwards, the call is routed back to network B towards UE-B. The related roaming interoperability configuration is named CFG_VoNR_RMI_2NW_B; where the suffix B signifies 'visited network B'.

6.1.6.2 Network components

The following components are required for this scenario:

One set each for PLMN-A and for PLMN B of the following:

- One 5G UE with IMS User Agents
- One gNB
- One 5GC with at least all of the following:
  - One AMF
  - One SMF
  - One UPF

  - One PCF
  - One N3IWF
- One WLAN
- One IMS with at least all of the following:
  - One P-CSCF
  - One I-CSCF/S-CSCF
  - One HSS
  - One AS (e.g. MMTel Application Server)
  - One IMS AGW/TrGW
  - One IBCF

6.1.6.3 Protocols and Reference Points

The following references will be interoperated in this scenario.

Interface	Protocol
N1/N2	5G NAS/NGAP
N5	HTTP2
Rx	Diameter
Gm	IMS SIP
Ic	IMS SIP

NOTE: One of N5 or Rx should be supported for this configuration.

6.2 IoT Configurations

6.2.1 IoT 5G SA

6.2.1.1 Configuration CFG_5GSA

The architecture for the 5GC related test case is shown in Figure 8.

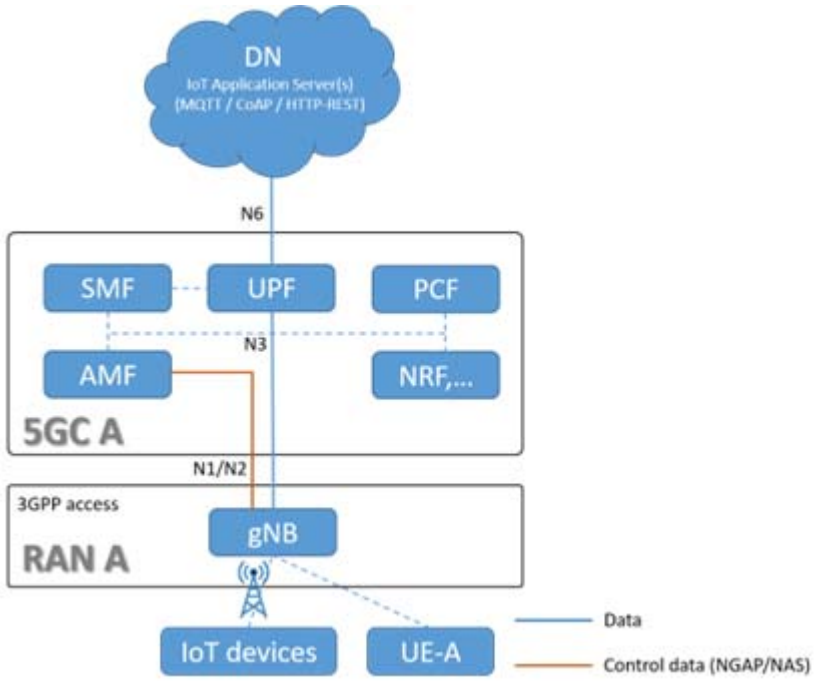


Figure 8: Configuration CFG_5GSA

Configuration CFG_5GSA is used for one peer network where users or IoT devices are 5G registered to their home network. UE-A or IoT device connects to network A represented by 5GC A and accesses IoT application services located in the Data Network (DN) via the UPF over the N6 interface. 5G Registration, 5G Deregistration signalling procedures of each user or IoT device are performed in this network. The DN may host IoT Application Server(s) supporting MQTT, CoAP and HTTP/REST based services.

6.2.1.2 Network components

The following components are required for this scenario:

- 5G UE, IoT devices, etc.
- One gNB
- One 5GC with at least all of the following:
  - One AMF
  - One SMF
  - One UPF
  - One PCF
- IoT Application Server(s) in the DN (e.g. MQTT Broker, CoAP Server, HTTP/REST Server)

6.2.1.3 Protocols and Reference Points

The following references will be interoperated in this scenario.

Interface	Protocol
N1/N2	5G NAS/NGAP
N3	GTP+Data
N6	IP-based IoT application traffic

6.2.2 IoT 5G SA Private to Public

6.2.2.1 Configuration CFG_5GSA_SNPN_PLMN

The architecture for the 5GC related test case is shown in Figure 9.

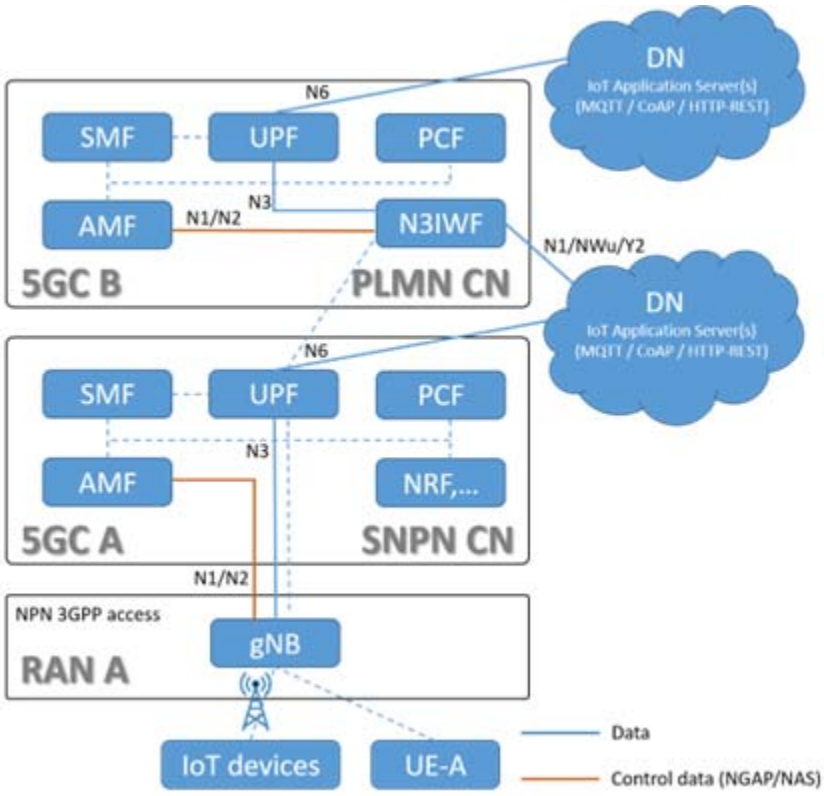


Figure 9: Configuration CFG_5GSA_SNPN_PLMN

Configuration CFG_5GSA_PLMN_SNPN is used for two peer networks where users or IoT devices are 5G registered to their public network. UE-A or IoT device connects to network A represented by 5GC A (PLMN CA) and accesses IoT application services located in the Data Network (DN) via the UPF over the N6 interface. 5G Registration, 5G Deregistration signalling procedures of each user or IoT device are performed in public network. UE-A or IoT device connects to network B represented by 5GC B (SNPN CN) and accesses IoT application services located in the Data Network (DN) via the UPF over the N6 interface. 5G Registration, 5G Deregistration signalling procedures of each user or IoT device can be performed in private network too for specific service. Each DN may host IoT Application Server(s) supporting MQTT, CoAP and HTTP/REST based services.

6.2.2.2 Network components

The following components are required for this scenario:

- 5G UE, IoT devices, etc.
- One gNB
- Two 5GC with at least all of the following:
  - One AMF
  - One SMF
  - One UPF
  - One PCF

- One N3IWF

• IoT Application Server(s) in the DN (e.g. MQTT Broker, CoAP Server, HTTP/REST Server)

6.2.2.3 Protocols and Reference Points

The following references will be interoperated in this scenario.

Interface	Protocol
N1/N2	5G NAS/NGAP
N3	GTP+Data
N6	IP-based IoT application traffic

6.2.3 IoT 5G SA Public to Private

6.2.3.1 Configuration CFG_5GSA_PLMN_SNPN

The architecture for the 5GC related test case is shown in Figure 10.

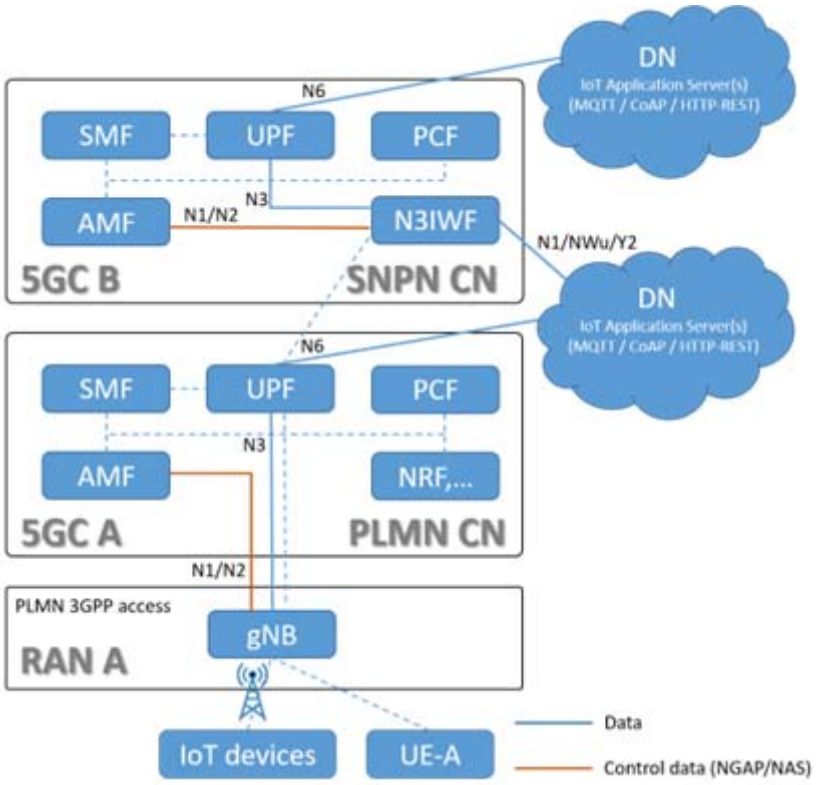


Figure 10: Configuration CFG_5GSA_PLMN_SNPN

Configuration CFG_5GSA_PLMN_SNPN is used for two peer networks where users or IoT devices are 5G registered to their public network. UE-A or IoT device connects to network A represented by 5GC A (PLMN CA) and accesses IoT application services located in the Data Network (DN) via the UPF over the N6 interface. 5G Registration, 5G Deregistration signalling procedures of each user or IoT device are performed in public network. UE-A or IoT device connects to network B represented by 5GC B (SNPN CN) and accesses IoT application services located in the Data Network (DN) via the UPF over the N6 interface. 5G Registration, 5G Deregistration signalling procedures of each user or IoT device can be performed in private network too for specific service. Each DN may host IoT Application Server(s) supporting MQTT, CoAP and HTTP/REST based services.

6.2.3.2 Network components

The following components are required for this scenario:

- 5G UE, IoT devices, etc.
- One gNB
- Two 5GC with at least all of the following:
  - One AMF
  - One SMF
  - One UPF
  - One PCF
  - One N3IWF
- IoT Application Server(s) in the DN (e.g. MQTT Broker, CoAP Server, HTTP/REST Server)

6.2.3.3 Protocols and Reference Points

The following references will be interoperated in this scenario.

Interface	Protocol
N1/N2	5G NAS/NGAP
N3	GTP+Data
N6	IP-based IoT application traffic

7 Interoperable Functions Statement (IFS)

7.1 VoNR UE Features

Table 4: VoNR UE features

Item	Entity	IFS_ID	Reference	Status	Support
1	Does the UE support IMS voice over NR connected 5GCN	UE_ims_voice_NR_5GCN	ETSI TS 124 501 [30], clause 4.3.2 and ETSI TS 138 306 [33], clause 4.2.13	m	
2	Does the UE support N3 data transfer	UE_N3_data	ETSI TS 124 501 [30], clause 9.11.3.1	m	

7.2 VoNR 5GC Features

Table 5: VoNR 5GC features

Item	Entity	IFS_ID	Reference	Status	Support
1	Does the 5GS support IMS Voice over PS Session over 3GPP access indicator?	5GS_ims_VoPS_3GPP	ETSI TS 124 501 [30], clause 9.11.3.5	m	
2	Does the 5GS support N3 data interface?	5GS_N3_data	ETSI TS 124 501 [30], clause 9.11.3.5	m	
3	Does the 5GS support Rx interface in the PCF?	5GS_PCF_Rx	ETSI TS 123 228 [23], clause 4.6.1.	o.1	
4	Does the 5GS support N5 interface in the PCF?	5GS_PCF_N5	ETSI TS 123 228 [23], clause 4.6.1.	o.1	
o.1: At least one of these options shall be supported.					

7.3 VoNR IMS Features

Table 6: VoNR IMS features

Item	Entity	IFS_ID	Reference	Status	Support
1	Does the IMS support Rx interface in the P-CSCF?	IMS_PCSCF_Rx	ETSI TS 123 228 [23], clause 4.6.1.	o.1	
2	Does the IMS support N5 interface in the P-CSCF?	IMS_PCSCF_N5	ETSI TS 123 228 [23], clause 4.6.1.	o.1	
o.1: At least one of these options shall be supported.					

7.4 IoT UE Features

Table 7: IoT UE features

Item	Entity	IFS_ID	Reference	Status	Support
1	UE supports N3 data transfer	UE_N3_data	ETSI TS 124 501 [30], clause 9.11.3.1	m	
2	UE supports control plane CIoT 5GS optimization	UE_CP_CIoT	ETSI TS 124 501 [30], clauses 5.3.21 and 9.11.3.1	o	
3	UE supports IP header compression for control plane CIoT 5GS optimization	UE_IPHC_CP_CIoT	ETSI TS 124 501 [30], clauses 5.3.21 and 9.11.3.1	o	
4	UE supports user plane CIoT 5GS optimization	UE_UP_CIoT	ETSI TS 124 501 [30], clauses 5.3.21 and 9.11.3.1	o	
5	UE supports Ethernet header compression for control plane CIoT 5GS optimization	UE_EHC_CP_CIoT	ETSI TS 124 501 [30], clauses 5.3.21 and 9.11.3.1	o	

7.5 IoT 5GC features

Table 8: IoT 5GC features

Item	Entity	IFS_ID	Reference	Status	Support
1	5GS supports N3 data interface	5GS_N3_data	ETSI TS 124 501 [30] clause 9.11.3.5	m	
2	5GS supports control plane CIoT 5GS optimization	5GS_CP_CIoT	ETSI TS 124 501 [30] clauses 5.3.21, 9.11.3.5	o	
3	5GS supports IP header compression for control plane CIoT 5GS optimization	5GS_IPHC_CP_CIoT	ETSI TS 124 501 [30] clauses 5.3.21 and 9.11.3.5	o	
4	5GS supports user plane CIoT 5GS optimization	5GS_UP_CIoT	ETSI TS 124 501 [30] clauses 5.3.21 and 9.11.3.5	o	
5	5GS supports Ethernet header compression for control plane CIoT 5GS optimization	5GS_EHC_CP_CIoT	ETSI TS 124 501 [30] clauses 5.3.21 and 9.11.3.5	o	

8 Signalling call flows

8.1 Signalling call flows for voice support

8.1.1 UE 5G registration for VoNR support

Each UE should register to the 5G network by performing the 5GS registration procedure as shown in Figure 11.

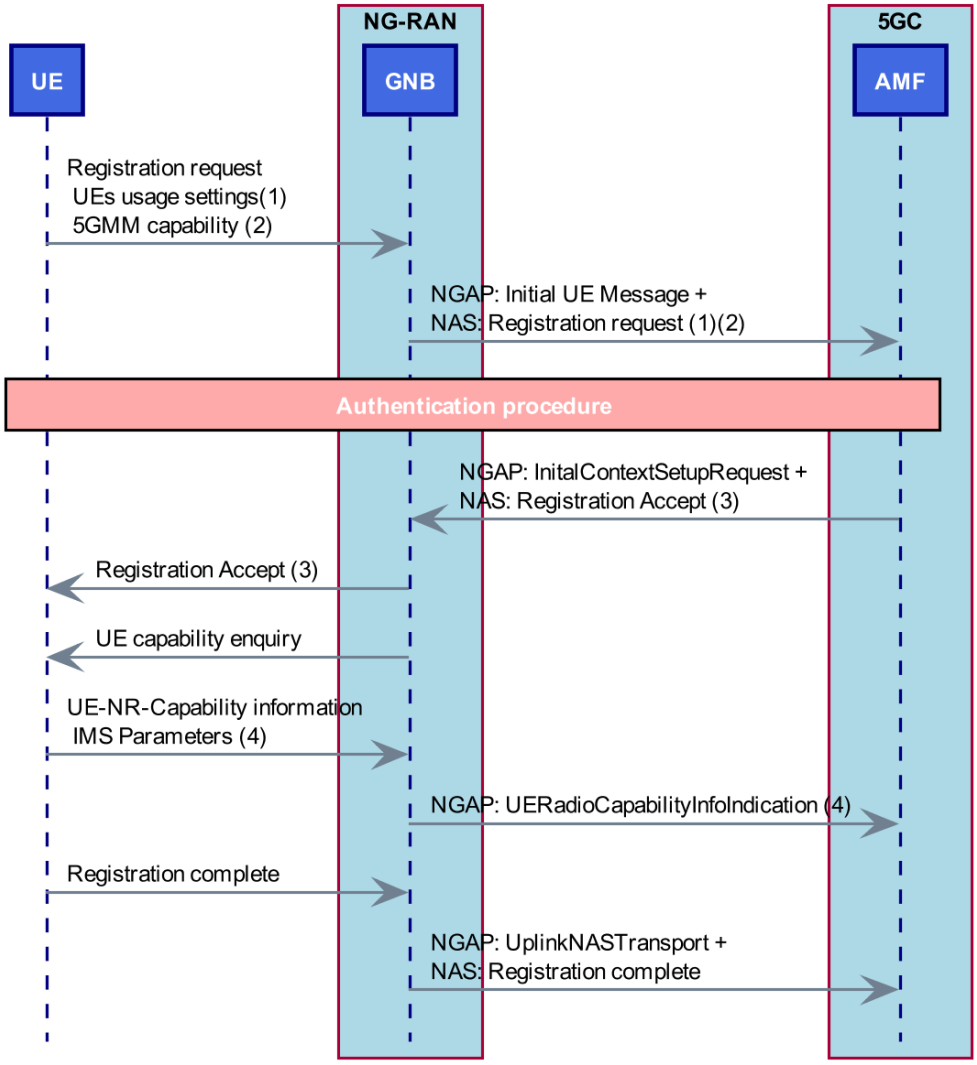


Figure 11: UE 5G Registration procedure

Table 9

VoNR 5G Registration Profile	
Use Cases for Voice Centric and Single-Registration Mode for originating and terminating VoNR calls	
Expected Behaviour	UE/Network should prioritize voice services for 5GS over NR-connected 5GCN. During the 5G registration procedure: • the UE indicates voice-centric usage and support for IMS voice over PS session over 3GPP access; • the UE indicates support for N3 user plane data transfer; • the 5GS confirms support for IMS Voice over PS session over 3GPP access; • no EPS fallback or interworking mechanisms (e.g. N26) are required.

VoNR 5G Registration Profile	
Network Feature Support IMS/VoNR	NR connected to 5GC (GSMA Option#2 [34]); Supports VoNR
Registration Request/UE's Usage Setting See note 1 ETSI TS 124 501 [30], clause 9.11.3.55	**Voice Centric**
Registration Request/ 5GMM capability See note 2 ETSI TS 124 501 [30], clause 9.11.3.1	S1 mode = NOT APPLICABLE (EPC NAS supported) N3 data = SUPPORTED (N3 data transfer)
Registration Accept/5GS network feature support See note 3 ETSI TS 124 501 [30], clause 9.11.3.5	• IMS-VoPS-3GPP = SUPPORTED (IMS Voice over PS Session over 3GPP access indicator) • IMS-VoPS-N3GPP = NOT APPLICABLE (IMS Voice over PS Session over Non-3GPP access indicator) • IWK N26 = NOT APPLICABLE (Interworking without N26 interface indicator) • N3 data = SUPPORTED
UE-NR-Capability (RRC) See note 4 ETSI TS 138 306 [33], clause 4.2.13 IMS parameters	UE supports IMS voice over NR connected 5GCN = TRUE. UE supports IMS voice over E-UTRA connected to 5GCN = NOT APPLICABLE. voiceOverSCG-BearerEUTRA-5GC= NOT APPLICABLE. FallbackIndicationEPS = NOT APPLICABLE
NOTE 1: See message (1) in Figure 11. NOTE 2: See message (2) in Figure 11. NOTE 3: See message (3) in Figure 11. NOTE 4: See message (4) in Figure 11.	

8.1.2 PDU session procedures for VoNR services

8.1.2.1 Scope and objective

PDU session procedures are required to support Voice over NR (VoNR) services in a 5G Standalone system. Establishment of a PDU session with a default QoS flow is used for IMS signalling. Subsequent PDU session modification is used to add a dedicated QoS flow for VoNR media.

8.1.2.2 PDU session establishment with default QoS flow

After successful 5G registration (see clause 8.1.1), the UE establishes a PDU session towards the data network supporting IMS connectivity as shown in Figure 12.

The PDU Session Establishment procedure results in the creation of one PDU session and one default QoS flow, which is always non-GBR. This default QoS flow is used for IMS registration signalling (SIP REGISTER, authentication) and IMS session control signalling prior to voice media activation.

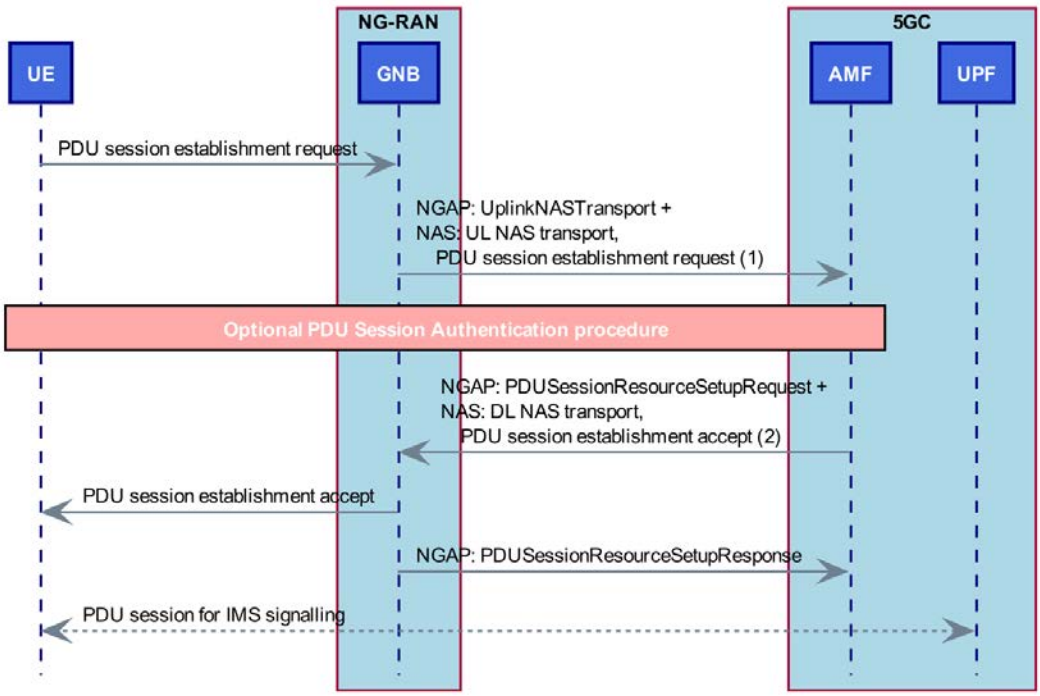


Figure 12: UE PDU establishment procedure for IMS signalling

Table 10

VoNR PDU Session Establishment Profile	
Use Cases for PDU session establishment with a default QoS flow for IMS signalling in Voice Centric and Single-Registration Mode for originating and terminating VoNR calls	
Expected Behaviour	During the PDU Session Establishment procedure: a PDU session is established towards the IMS data network (DNN); a default QoS flow is assigned and associated with the PDU session; the default QoS flow is of resource type Non-GBR; IMS signalling traffic is mapped to the default QoS flow via QoS Rules; the PDU session remains active after IMS registration and throughout the VoNR call lifecycle. The default QoS flow carries IMS signalling and control-related traffic and does not carry conversational voice media.
PDU Session Establishment Request ETSI TS 124 501 [30], clause 8.3.1	PDU Session Type: IPv4, IPv6 or IPv4v6 SSC mode: SSC mode 1 (typical for VoNR) DNN: "ims" or operator-defined IMS data network S-NSSAI: • SST=1 (eMBB option) • SD: operator-defined
PDU Session Establishment Accept ETSI TS 124 501 [30], clause 8.3.2	PDU Session ID QoS Rules: default QoS rule QoS Flow Descriptions: • Default QoS Flow Identifier (QFI) assigned by the network • 5QI = 5 (IMS signalling as non-GBR due to ETSI TS 123 501 [26], Table 5.7.4-1)

8.1.2.3 PDU session modification for VoNR dedicated QoS flow

When a VoNR call is initiated or terminated, IMS signalling triggers QoS policy control in the 5GS, resulting in a PDU Session Modification procedure as shown in Figure 13. This procedure adds a dedicated QoS flow to the already established PDU session. The dedicated QoS flow is used exclusively for VoNR media traffic (RTP/RTCP).

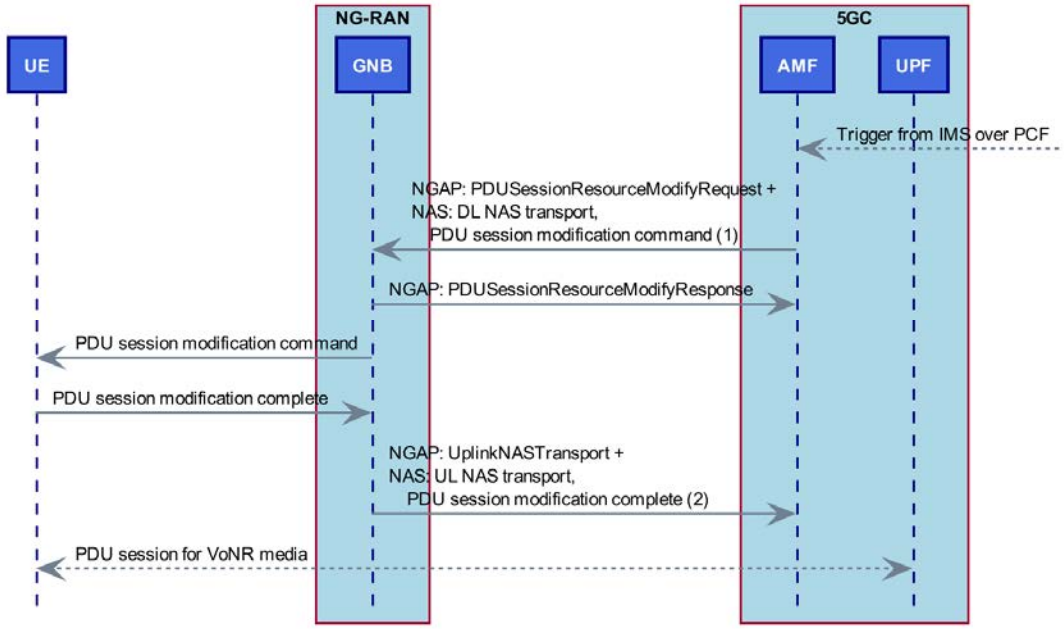


Figure 13: Network PDU modification procedure for VoNR media

Table 11

VoNR PDU Session Modification Profile	
Use Cases for PDU Session Modification to add a dedicated QoS flow for conversational voice media in Voice Centric and Single-Registration Mode for originating and terminating VoNR calls	
Expected Behaviour	Upon VoNR call establishment: • a PDU Session Modification procedure is triggered; • a dedicated QoS flow is added to the existing PDU session; • the dedicated QoS flow is associated with a voice-optimized standardized 5QI (e.g. 5QI = 1); • the dedicated QoS flow is of resource type GBR; • conversational voice media (RTP/RTCP) traffic is mapped to the dedicated QoS flow via QoS Rules; • the dedicated QoS flow remains active for the duration of the VoNR session; • upon call release, the dedicated QoS flow is removed while the default QoS flow and PDU session remain established. Conversational voice media is not transported over the default QoS flow.
PDU Session Modification Command ETSI TS 124 501 [30], clause 8.3.9	QoS Rules: • New QoS rule mapping VoNR media traffic to a dedicated QoS flow • QoS Flow Descriptions: • Dedicated QoS Flow Identifier (QFI) assigned by the network • 5QI = 1 (Conversational Voice as GBR due to ETSI TS 123 501 [26], Table 5.7.4-1) • GFBR uplink • GFBR downlink
PDU Session Modification Complete ETSI TS 124 501 [30], clause 8.3.10	UE confirms acceptance of the dedicated QoS flow and applies the updated QoS mapping

8.1.3 UE IMS registration

This clause describes the IMS registration procedure applicable to Voice Centric and Single-Registration Mode for originating and terminating VoNR calls in a 5G Standalone system.

After successful 5G registration and establishment of a PDU session with a default QoS flow supporting IMS signalling, the UE performs IMS registration using SIP signalling towards the IMS core network as shown in Figure 14. IMS registration is transported over the default QoS flow within the established PDU session. The procedure includes transmission of a SIP REGISTER request from the UE towards the P-CSCF, authentication of the UE by the IMS core, and confirmation of successful registration.

Upon successful completion of the procedure, the UE is registered in the IMS core network and the service profile associated with the subscriber is activated. The UE is able to originate and terminate VoNR calls.

IMS registration is logically independent from the subsequent addition of a dedicated GBR QoS flow for conversational media.

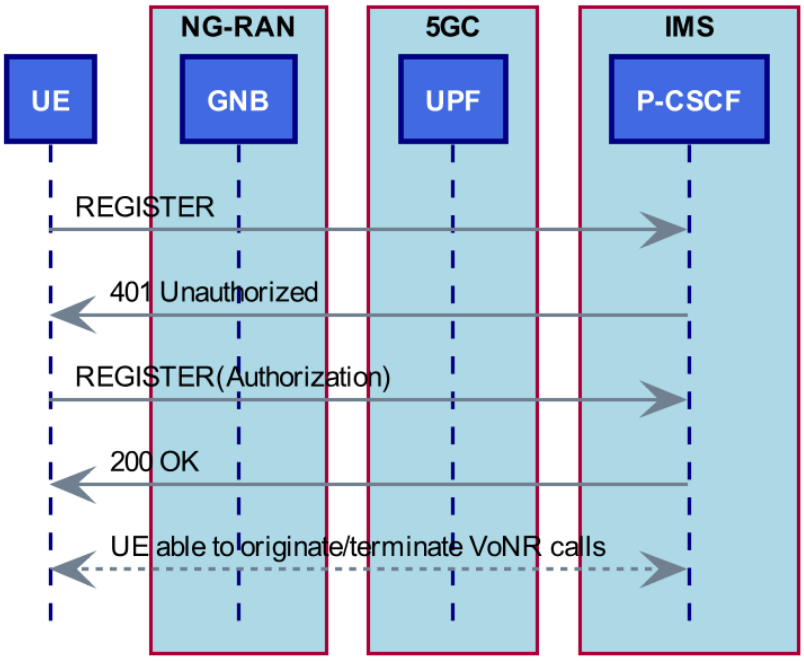


Figure 14: IMS registration procedure over NR-connected 5GC

8.1.4 VoNR Call Establishment between two UEs

This clause describes the call establishment procedure applicable to Voice Centric and Single-Registration Mode for originating and terminating VoNR calls between UE-A and UE-B in a 5G Standalone system.

After successful 5G registration, establishment of a PDU session with a default QoS flow supporting IMS signalling, and IMS registration of both UEs, a VoNR call is established using SIP signalling within the IMS core network as shown in Figure 15. The procedure is initiated by transmission of a SIP INVITE request from the originating UE (UE-A) towards the IMS core and routed to the terminating UE (UE-B). The signalling exchange enables session establishment and negotiation of media parameters.

IMS signalling related to call establishment is transported over the default QoS flow within the established PDU session. Upon successful session establishment, the network initiates a PDU Session Modification procedure for both UEs to add a dedicated QoS flow for conversational voice media. Conversational voice media (RTP/RTCP) is transported over the dedicated GBR QoS flow within the established PDU session.

Successful completion of the procedure results in an active IMS session between UE-A and UE-B and enables bidirectional conversational voice communication over NR-connected 5GC.

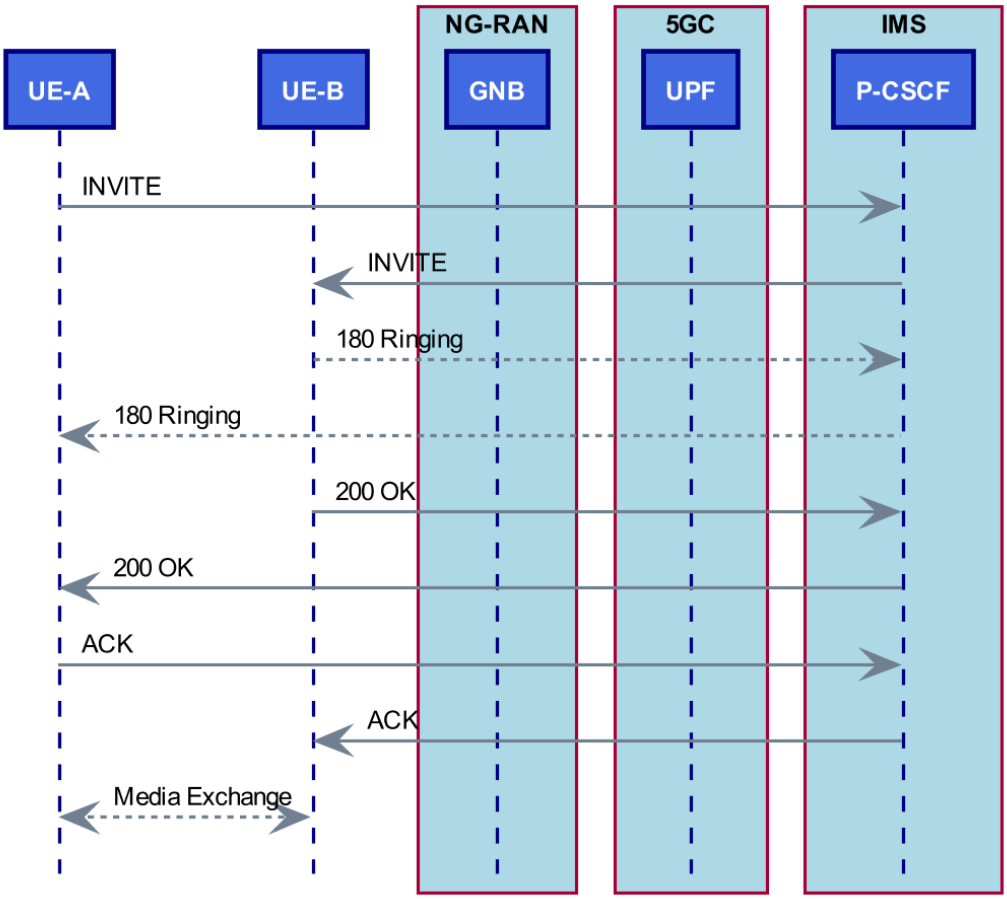


Figure 15: Call establishment between two UEs

8.2 Signalling call flows for IoT support

8.2.1 IoT UE 5G registration

Each IoT device should register to the 5G network by performing the 5GS registration procedure as shown in Figure 16.

IoT devices may perform an initial 5GS registration procedure or subsequent mobility and periodic registration update procedures.

During the initial registration procedure, authentication and security establishment procedures are typically performed between the UE and the 5GS.

For subsequent registration update procedures, the network may reuse an existing valid security and mobility management context. In such cases, authentication and security procedures may be reduced or omitted in accordance with the standard 5GS procedures defined in ETSI TS 123 502 [27] and ETSI TS 124 501 [30].

These optimizations reduce signalling overhead and improve power efficiency for IoT-oriented deployments while remaining within the standard 5GS registration procedure.

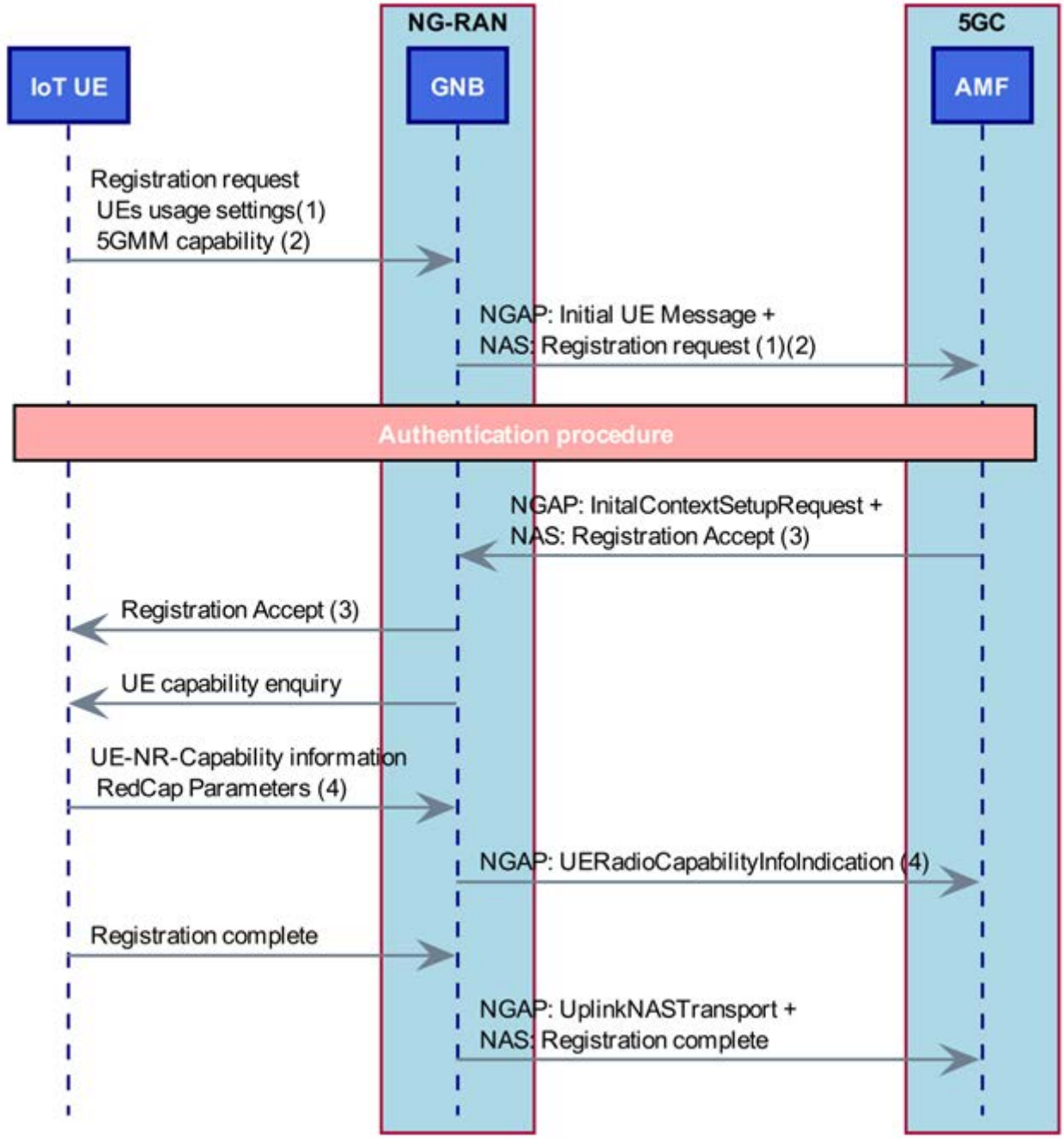


Figure 16: IoT UE 5G Registration procedure

Table 12

IoT 5G Registration Profile	
Registration profile for data-centric IoT devices accessing 5GS packet data services.	
Expected Behaviour	UE/Network should support data-centric IoT operation over 5GS. Upon successful registration: • the UE and 5GS successfully complete the registration procedure; • the UE is authorized to access IoT packet data services; • the UE usage setting is recognized as data-centric; • CIoT optimizations may be applied where supported; • the UE capability profile for IoT operation is accepted by the network.
Network Feature Support IoT	Support of data-centric IoT operation over 5GS including: • CIoT optimization support; • RedCap UE support where applicable; • extended DRX (eDRX) support; • small data transmission support.
Registration Request/ UE's Usage Setting See note 1 ETSI TS 124 501 [30], clause 9.11.3.55	**Data Centric**
Registration Request/ 5GMM Capability IE See note 2 ETSI TS 124 501 [30], clause 9.11.3.1	S1 mode = NOT APPLICABLE (EPC NAS supported). N3 data = SUPPORTED (N3 data transfer). 5G-CP CIoT = OPTIONAL (Control plane CIoT 5GS optimization). 5G-IPHC-CP CIoT = OPTIONAL (IP header compression for control plane CIoT 5GS optimization). 5G-UP CIoT = OPTIONAL (User plane CIoT 5GS optimization). 5G-EHC-CP CIoT = OPTIONAL (Ethernet header compression for control plane CIoT 5GS optimization).
Registration Accept/5GS Network Feature Support IE See note 3 ETSI TS 124 501 [30], clause 9.11.3.5	IWKN26 = NOT APPLICABLE 5G-CP CIoT = OPTIONAL (Control plane CIoT 5GS optimization). 5G-IPHC-CP CIoT = OPTIONAL (IP header compression for control plane CIoT 5GS optimization) N3 data = SUPPORTED (N3 data transfer). 5G-UP CIoT = OPTIONAL (User plane CIoT 5GS optimization). 5G-EHC-CP CIoT = OPTIONAL (Ethernet header compression for control plane CIoT 5GS optimization).
UE-NR-Capability (RRC) See note 4 ETSI TS 138 306 [33], clause 4.2.21	Reduced Capability (RedCap) UE support = OPTIONAL.
NOTE 1: See message (1) in Figure 16. NOTE 2: See message (2) in Figure 16. NOTE 3: See message (3) in Figure 16. NOTE 4: See message (4) in Figure 16. NOTE 5: CIoT and RedCap related capabilities may reduce signalling overhead, UE complexity and power consumption for IoT-oriented deployments. Such optimizations are negotiated within above standard 5GS registration procedure through capability exchange and network feature support signalling and do not represent a separate 3GPP registration procedure.	

8.2.2 PDU session procedures for IoT

PDU session procedures for IoT services enable transport of IP-based application traffic between the IoT device and the application server over the 5G System using user plane connectivity and QoS flows as shown in Figure 17.

Establishment of a PDU session with configured QoS characteristics enables support of MQTT, CoAP and HTTP/REST IoT communication services and allows validation of QoS handling based on standardized 5QI characteristics.

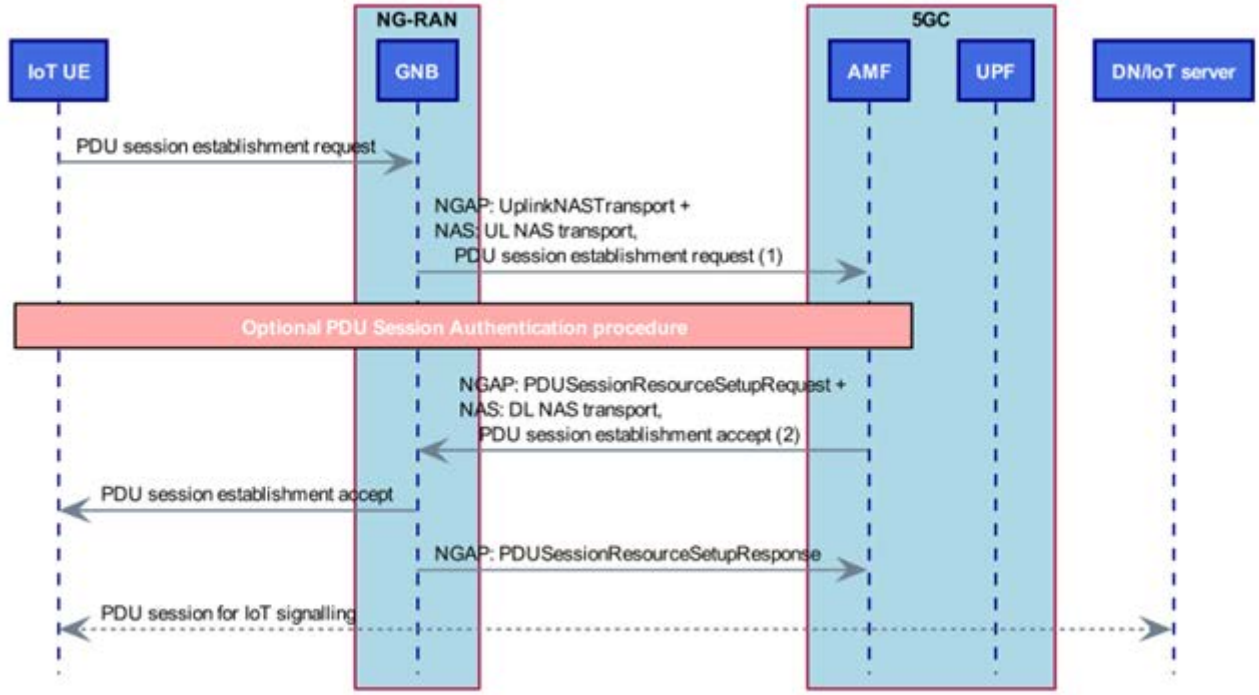


Figure 17: UE PDU establishment procedure for IoT data services

Table 13

IoT PDU Session Establishment Profile	
Profile for PDU Session establishment with a default QoS Flow for data-centric IoT devices over 5GS.	
Expected Behaviour	UE/Network should establish a PDU Session supporting IoT packet data services over 5GS. Upon successful PDU Session establishment: • the UE is assigned IP connectivity towards the Data Network; • a default QoS Flow is established; • the configured QoS characteristics are applied to IoT traffic; • the UE is able to exchange MQTT, CoAP and HTTP/REST application traffic; • the network applies QoS handling according to the configured QoS characteristics defined in clause 10 of present document.
PDU Session Establishment Request ETSI TS 124 501 [30], clause 8.3.1	Request type: INITIAL REQUEST PDU Session Type: IPv4, IPv6 or IPv4v6 SSC mode: SSC mode 1 DNN: "iot" or operator-defined data network S-NSSAI: • SST=3 (mMTC option) • SD: operator-defined Always-on PDU Session Requested = OPTIONAL
PDU Session Establishment Accept ETSI TS 124 501 [30], clause 8.3.2	PDU Session ID QoS Rules: Default QoS rule QoS Flow Descriptions: • Default QoS Flow Identifier (QFI) assigned by the network • 5QI assigned according to IoT service requirements (see clause 10.3) DNN accepted by the network S-NSSAI accepted by the network Session-AMBR assigned according to operator policy.
NOTE: Certain IoT deployments may use control plane CIoT 5GS optimization for small data transfer using NAS signalling without establishment of a user plane PDU session. Such deployments are outside the primary scope of QoS flow validation and end-to-end QoS testing described in clause 10 of the present document.	

9 QoS and QoE Tests for voice over NR

9.1 Introduction to the concept

Clause 9 specifies the framework for the evaluation of QoS and QoE for end-to-end voice connections provided over the 5G VoNR interoperability test configurations described in the present document.

Due to the concept of the present document to serve as a basis for the planning and preparation of test events, the description of the QoS related tests is considerably different to the approach that would be used for an equipment specification.

The objective of these tests is to verify that interoperable implementations of 5G network elements, terminals and IMS components are capable of establishing an end-to-end voice connection that meets the expected transmission quality from the perspective of the end user.

For equipment brought by manufacturers, network operators, vendors and other stakeholders, it is not possible to assume that equipment brought to test events complies to specific standards or contains certain hardware or software components.

Therefore, the QoS test description in the present document, in various places, provides optional approaches which may be chosen by the test house in charge; such choices have to be documented.

As a starting point, it is assumed that the test house in charge, is very well familiar with the relevant standards and that the procedures used to conduct the tests outlined in this clause comply with these standards to the utmost extend. Any deviation from these standards during the execution of the tests shall be justified and documented.

The evaluation performed in the present document focuses on the following aspects:

- verification that QoS parameters required for voice communication are correctly negotiated and applied;
- verification that the resulting end-to-end connection provides adequate speech transmission quality;
- objective estimation of perceived speech quality based on standardized measurement methods.

The tests described in this clause are intended for use during interoperability events and demonstrations where the equipment under test may originate from different vendors and may not necessarily comply with identical implementation profiles.

The procedures therefore focus on **end-to-end observable behaviour and measured media quality**, rather than detailed conformance to individual protocol specifications.

9.2 Reference network scheme

9.2.1 Overview

Figure 18 depicts the elements of the 5G network under test as seen from the perspective of QoS and QoE testing. Terminals, if available should be 5G capable, in the simulation box, packet loss and codec tandeming may be simulated as appropriate.

The QoS and QoE tests described in the present clause are conducted using a reference network configuration representing an end-to-end VoNR communication path between two user equipment.

The reference network shall include:

- two user equipments supporting VoNR;
- the NG-RAN access network;
- the 5G Core network;
- the IMS core providing voice services.

The measurement configuration shall allow the injection of speech test signals at one end of the communication path and the recording of the degraded signal at the receiving end.

Where required, controlled impairments may be introduced in order to analyse the behaviour of the system under defined transmission conditions.

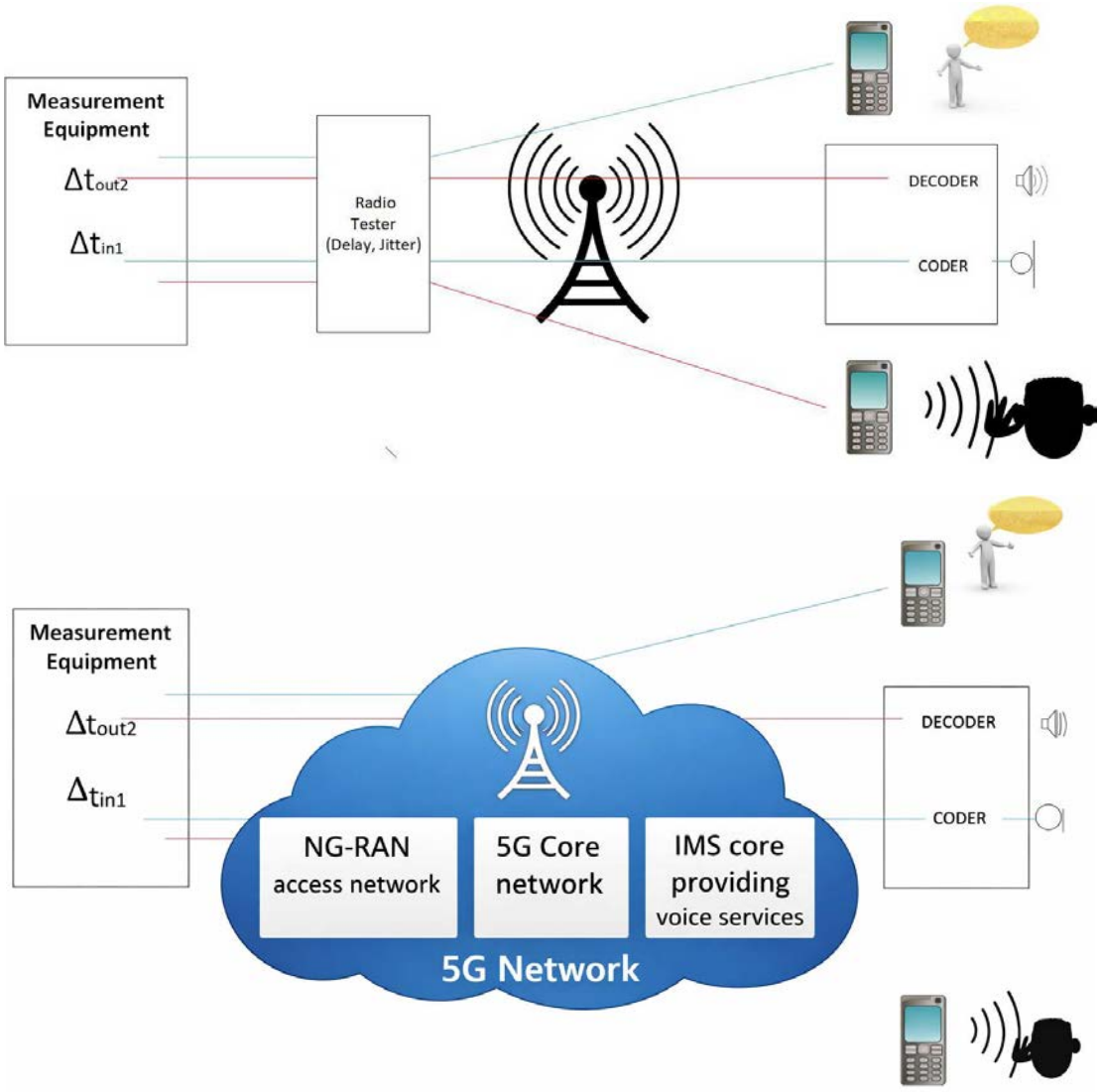


Figure 18: Elements of a 5G network from the perspective of QoS and QoE testing

For multi-rate codecs, like AMR or EVS the actual bitrate mode adjusted by the system internal QoS signalling mechanisms shall be recorded.

9.2.2 Simulation of packet loss or jitter

In case a wider range of QoS and QoE related network conditions is desired, packet loss or jitter may be simulated inside the 5G network under test.

The values for packet loss simulation or jitter simulation are to be chosen with view to realistic network conditions but not for codec qualification purposes.

Such impairments may be introduced by:

- network impairment simulation tools;
- configurable traffic shaping elements;
- controlled test network configurations.

Examples for similar network in a 4G environment are described in Recommendations ITU-T G.1028 and G.1028.2 (see bibliography).

The parameters used for impairment simulation shall be documented in the test report.

This is explicitly to be considered as optional.

9.2.3 Simulation of codec tandeming

In case a wider range of QoS and QoE related configurations is desired, codec tandeming may be simulated inside the 5G network under test.

Codec tandeming is avoided whenever possible because it deteriorates speech quality as perceived by the user (see bibliography, clause C.3). However codec tandeming will occur in certain cases of network interconnection, routing via OTT or in case of the use low quality user equipment.

This is explicitly to be considered as optional.

9.3 Injection and retrieval of speech test signals

9.3.1 Introduction

The measurement of end-to-end speech quality requires the injection of reference speech signals into the communication path and the recording of the degraded signal at the receiving side.

Two alternative methods as described in clauses 9.3.2 and 9.3.3 may be used. Each of these methods may be used at either end of the test setup.

Figure 19 through Figure 22 depict these options.

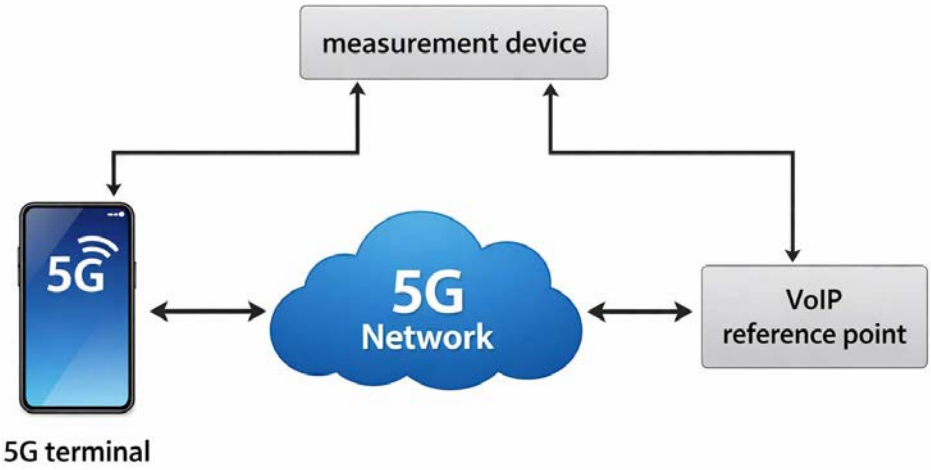


Figure 19: Terminal on A side and VoIP reference point on B side

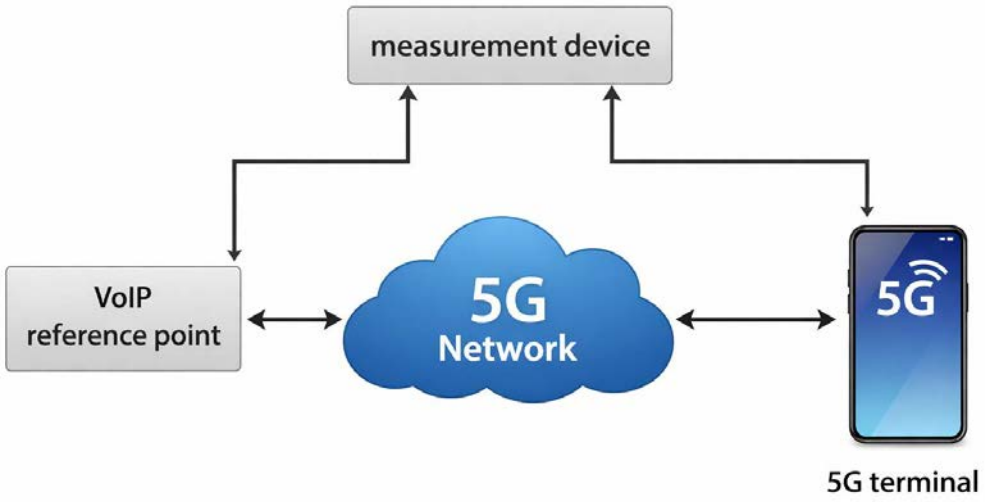


Figure 20: VoIP reference point on A side and Terminal on B side

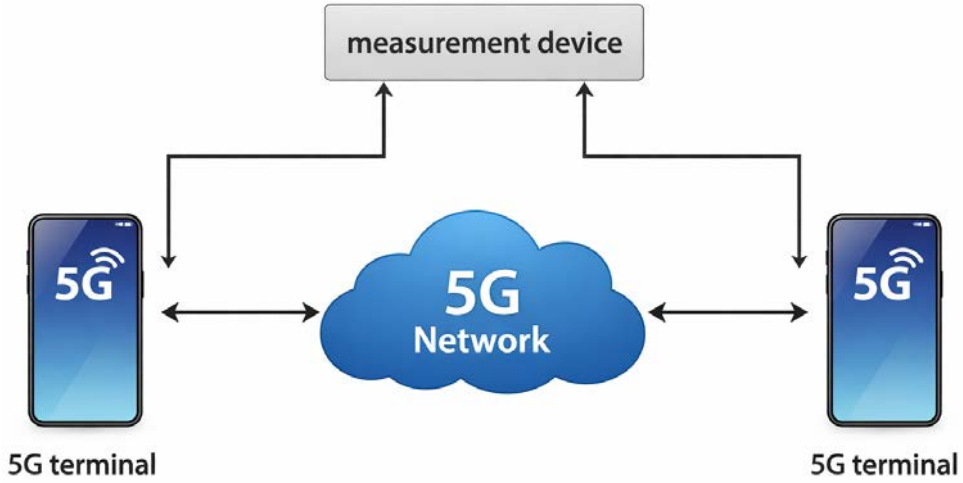


Figure 21: Terminals on either side

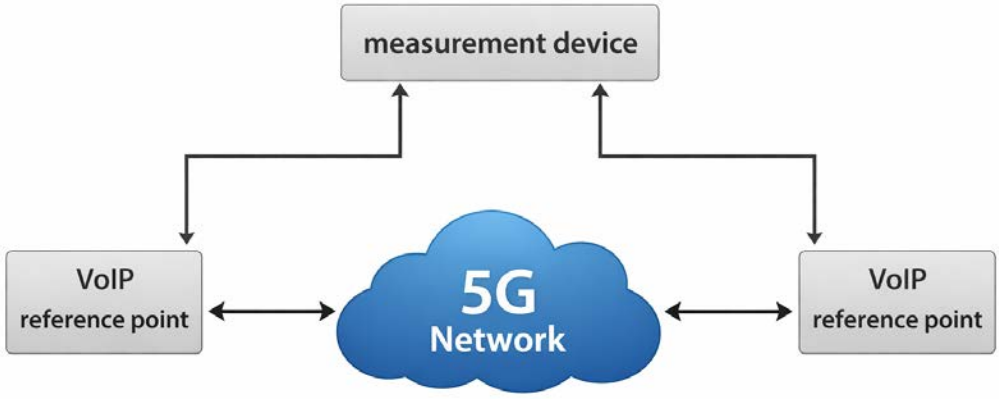


Figure 22: VoIP reference points on either side

9.3.2 Injection and retrieval via the acoustical interface of a 5G terminal

The most realistic assessment of end-to-end voice communication is - of course - the one via acoustical interfaces.

The test set-up for measurements between acoustic interface shall comply with clause 5.2 of ETSI TS 103 739 [18].

Due to the extra efforts in terms of both testing apparatus and testing environment, this option may be considered as less feasible for use during dedicated test events.

9.3.3 Injection and retrieval via an IP half channel adaptor

This requires the availability of a device compliant with clause 7 of ETSI TS 102 425 [16]:This type of test access includes the following options:

- electrical interfaces of a terminal;
- Bluetooth® interface of a terminal;
- digital interface of the network;
- digital software implementation in combination with the network.

The method chosen needs to be well documented in the test report of the event.

9.3.4 Requirements for speech test signals

Speech signals used for objective listening quality measurements shall comply with the requirements defined in Recommendation ITU-T P.863 [12] and Recommendation ITU-T P.863.1 [13].

Reference speech signals should satisfy the following conditions:

- at least **3 s of active speech**;
- no more than **6 s of continuous active speech**;
- silence intervals of at least **500 ms** between speech segments;
- total sample duration not exceeding **12 s**;
- active speech level approximately **-26 dBov**;
- **16-bit linear PCM encoding**;
- background noise level below **-75 dBov (A)**.

To obtain representative results, the speech material should include multiple speakers.

It is recommended that the test set include speech samples from at least:

- two male speakers; and
- two female speakers.

At least two samples per speaker should be used.

In addition, the transparency requirement of the algorithm according to Recommendation ITU-T P.863 [12] shall be fulfilled. This requirement can be satisfied by using the test sentences provided in Annex D of Recommendation ITU-T P.501 [9].

9.4 Selection of QoS parameters

9.4.1 General

The parameters listed in this clause have the potential to be tested during an interoperability test event which aims at QoS and QoE aspects. The assessment of End-to-End Listening Quality MOS-LQOF as per clause 9.4.2 is mandatory; all others are optional and are left to the discretion of the stakeholders. Table 14 summarizes the effort for testing the parameters discussed in this clause.

Table 14: Effort for testing QoS parameters

QoS parameter	test signal injection /retrieval	real time test required	terminal environment	remarks
MOS-LQOF	all options	no	all options	
Delay	all options	yes	all options	
Double Talk	acoustical	yes	advance acoustical	
Insertion Loss	acoustical	yes	advance acoustical	
Talker Echo	acoustical	yes	advance acoustical	

9.4.2 End-to-end listening quality MOS-LQOF

9.4.2.1 Assessment compliant with P.863

The assessment of Listening Quality applies to both directions of transmission.

Since 5G networks and terminals exchange implementations can safely be assumed to be capable of delivering end-to-end fullband voice services (or at least super-wideband), the requirements are assessed on the fullband scale for the objective prediction of Listening Quality (MOS-LQOF).

MOS-LQO shall be assessed in accordance with Recommendation ITU-T P.863 [12] in conjunction with its application guide Recommendation ITU-T P.863.1 [13]. The algorithm of Recommendation ITU-T P.863 [12] shall be used in the fullband mode, only.

NOTE 1: P.863 provides the description of one validated algorithm, only. Beside the fullband mode, there exists a narrowband mode which is simplified and allows retrospective comparison with older test result obtained with the predecessor of the P.863 algorithm. The ITU-T does not recommend the use of this retrospective mode for any current applications.

NOTE 2: The main body of P.863 contains the functional description of the algorithm, whereas, annex B contains the pro forma C code which contains detailed guidance on the implementation details.

NOTE 3: For the purpose of the present document appropriate guidance can be found in the bibliography.

NOTE 4: Recommendation ITU-T P.863 [12] in fullband mode takes into account bandwidth limitations by detecting the absence of any speech energy above 3,8 kHz to indicate a narrowband degraded file and the absence of any speech energy above 7 kHz to indicate a wideband degraded file. With a narrowband signal the maximum achievable score is 3,8. The scale for MOS therefore is always from 1 to 5.

NOTE 5: Recommendation ITU-T P.863 [12] in fullband mode takes into account changes in speech signal level as transmission degradations.

The transparent speech files provided in Recommendation ITU-T P.501 [9], Annex C or their composition in Recommendation ITU-T P.501 [9], Annex D shall be used for the tests.

The test shall be conducted at least with speech files in 2 languages. Male and female speech files should be used in a balanced matter which also allows for a gender specific post-analysis of the test results.

No pre-filtering shall be applied, and no level adaptation of the received signals shall be done.

9.4.2.2 Assessment compliant with P.565.1

In cases which are compliant with the scope of Recommendation ITU-T P.565.1 [10] (e.g. that there is pure IMS calling or connections are used without transcoding or re-packaging, etc.), MOS-LQOF can alternatively be obtained by following Recommendation ITU-T P.565.1 [10]. This shall be stated in the test report.

9.4.2.3 Assessment according to proprietary methods

In cases where proprietary methods to assess MOS-LQOF are planned to be used, this can be done in addition to the Recommendation ITU-T P.863 [12] assessment according to clause 9.4.2.1 of the present document.

Details have to be recorded in the test report and it is also suggested to provide a statistical comparison analysis between results according to both methods.

9.4.3 Mean one-way delay of speech signals

Mean One-Way Delay (MOWD) is the average time it takes for a speech signal to travel from a source device to a destination device across a network.

For that purpose, one-way delay is tested separately in both directions of transmission in accordance with clause 6.12 of ETSI TS 103 739 [18].

Mean one-way delay is derived by averaging the one-way delay values for both directions of transmission.

The delay values per direction can also be derived from the assessment of MOS-LQOF by using the currently licensed implementation of Recommendation ITU-T P.863 [12].

NOTE: Caution has to be applied to mean one-way delay values obtained in test scenarios with virtual test set-ups, i.e. in cases where the equipment under test is not located physically in the same place.

9.4.4 Double Talk capability

Double talk capability shall be assessed in accordance with the methodology described in clause 6.8 of ETSI TS 103 739 [18].

As far as no unwanted signal processing impairments occur in the network section of the end-to-end voice communication path, this test will only reveal the combined properties of the voice terminals at both ends of the connection.

NOTE: Network mis-behaviour resulting in DSP overload impairments such as "digital crosstalk" may occur in cases where additional background load is applied to the network.

9.4.5 Insertion loss

No insertion loss is expected in fully digital implementations. However, the existence of insertion loss had been observed in faulty implementations in the past.

Therefore, this measurement serves as a pure sanity check.

Insertion loss can be derived by obtaining in a first step the RLR and the SLR as described in clause 6.2 of ETSI TS 103 739 [18] for each direction of transmission.

In a second step the end-to-end loudness rating OLR is determined following the same measurement methodology, for each direction of transmission.

The insertion loss Li is finally determined as the difference:

$$Li = OLR - RLR - SLR$$

for each direction of transmission.

9.4.6 Talker echo

Talker echo shall be assessed in accordance with the methodology described for terminal coupling loss in clause 6.7 of ETSI TS 103 739 [18].

As far as no unwanted signal processing impairments occur in the network section of the end-to-end voice communication path, this test will only reveal the combined properties of the voice terminals at both ends of the connection.

NOTE: Network mis-behaviour resulting in DSP overload impairments such as "digital crosstalk" may occur in cases where additional background load is applied to the network.

9.5 Statistical analysis and reporting

9.5.1 General

The raw data should be uploaded to a dedicated server of the test event organizer, in order to have them available for any kind of post-processing. Raw data are typically the property of the test event organizer.

The statistical analysis and the report shall be done in compliance with the principles and methodologies outlined in ETSI TS 102 250-6 [17] and Recommendation ITU-T P.1401 [15] - as far as applicable.

NOTE: The recommendations given in clause A.1.1 of ETSI TS 102 250-6 [17] with respect to use of statistical software, the limitations of MS Excel and the preferred use of the open source language and environment R (see http://www.r-project.org/) should be taken into account.

9.5.2 Raw data

Raw data typically include, but are not limited to:

- Configuration identifiers (anonymous IDs).
- Test condition identifiers (codec, network impairment, system under test).
- Individual predictions (e.g. 1 - 5 MOS scale).
- Session metadata (device, environment, system internal codec and bitrate selection).
- Recorded speech samples.

9.5.3 Data cleansing, outlier removal and expert listening

It is recommended that prior to analysis, an inspection of the objective test data be performed to uncover any types of outliers.

A typical example is the time clipping effect, often seen in speech quality metrics. Severe time clipping conditions can generate a fairly large difference between the Absolute Category Rating (ACR) listening-only subjective scores and the objective scores that rely on the comparison between the original and the degraded speech samples. It could be desirable to evaluate these types of situations within specially designed experiments, or to remove these outliers from the evaluation.

In addition, conditions or databases showing scores with unexpected confidence intervals, as compared with other conditions or databases obtained in experiments with similar designs, should be carefully analysed and a decision made on how these will be used in the reporting process. Likewise, similar databases which show extremely different scores for the anchor samples should be carefully considered in the evaluation.

9.5.4 Reporting and documentation

Final reports should include:

- Number of discarded results.
- MOS with confidence intervals.
- Statistical test methods.
- Any normalization applied.

This ensures reproducibility and credibility, especially for standardization or regulatory use.

9.5.5 Presentation of selected results

For showcase presentations of the test event results a representative summary should be prepared.

9.5.6 Specific recordings of oddities for presentation purposes

Any outstanding results - both extremely good quality and extremely odd impairments - should be prepared for presentation, also with audio samples.

9.6 Classification of results

9.6.1 Introduction

Due to the current lack of genuine benchmarks, the following subclauses provide known anchor points for each of the QoS parameters discussed in the present document.

9.6.2 MOS-LQOF

Since MOS-LQOF is an objective prediction methodology of MOS-LQSF, which in turn can only be obtained from auditory (subjective) test which follow the principles of Recommendation ITU-T P.800 [11], a concise and detailed benchmark data base for 5G MOS-LQOF is currently not available.

Therefore, it is suggested to classify results obtained during test events according to one or more of the following three approaches:

1) Anchor points coming from the standardization of P.863 itself. Those are:

   - MOS-LQOF = 3,8 for a narrowband connection (typically constituted by Recommendation ITU-T G.711 [7] coding) with no impairments other than the limitation in bandwidth added.

   - MOS-LQOF = 4.8 for a fullband connection with satisfies the transparency requirement of Recommendation ITU-T P.863.1 [13].

2) Appendix II of Recommendation ITU-T P.501 [9] provides typical results for a number of other encodings.

NOTE 1: The results in Appendix II of Recommendation ITU-T P.501 [9] provide results for codec reference implementation mostly under lab conditions - operational conditions may vary.

3) Anchor points for MOS-LQOF shall also be calculated by using a simplified formula (7.1) from Recommendation ITU-T G.107.2 [3] as follows:

   Original Recommendation ITU-T G.107.2 [3] formula:

$$R = Ro, FB - Is, FB - Id, FB - Ie, eff, FB + A \quad (1)$$

   For the context of the present document, it can be assumed:

   Ro,FB = 148

   Is,FB = 0

   Id,FB=0 and

   A=0

   Therefore, the simplified G.107.2 formula reads as follows:

$$R = 148 - 0 - 0 - Ie, eff, FB + 0 \quad (2)$$

   Ie,eff,FB can be determined as the sum of the fullband equipment impairment factors for all codecs concatenated in an end-to-end voice telephony connection. Ie values for a number of popular codecs can be found in Recommendation ITU-T G.113 [4] and shall be used.

   Equipment impairment factors provided for the narrowband E-model of Recommendation ITU-T G.107 [1] or for the wideband extension of the E-model of Recommendation ITU-T G.107.1 [2] shall be scale converted before they can be used in a fullband context. Instructions for the conversion in Recommendations ITU-T G.107.1 [2] and G.107.2 [3], respectively, shall be followed.

In order to transform the resulting R factor back to the MOS scale into a legitimate MOS-LQOF value, formulae A-1 and A-2 from Annex A of Recommendation ITU-T G.107.2 [3] can be used:

The transmission rating factor *R* can be in the range from 0 to 148, where *R* = 0 represents an extremely bad quality, and *R* = 148 represents a very high quality in the FB case. An estimated mean opinion score (MOS_{CQEF}) for the conversational situation on a scale of 1 - 5 can be obtained from the *R*-factor by using the equations:

$$Rx = \frac{R}{1.48} \quad (3)$$

For $Rx < 0$: $MOS_{CQEF} = 1$

For $0 < Rx < 100$: $MOS_{CQEF} = 1 + 0{,}035Rx + Rx(Rx - 60)(100 - Rx) \cdot 7 \cdot 10^{-6}$ (4)

For $Rx > 100$: $MOS_{CQEF} = 4{,}5$

For the purposes of the present document and hence for test events it is permissible to assume the postulate: MOS-LQOF: = MOS-LQEF.

NOTE 2: One of the obstacles of this approach is the mismatch of the upper scale bounds:

- 4,5 in the estimation model vs. 4,8 in the objective assessment methodology.

Table 15: Quick overview of Ie and MOS-LQEF in pure 5G networks (without transcoding):

	2G	3G	5G	Ie,nb	Ie,wb	Ie,fb	R,fb	MOS-LQE,fb
NB	GSM-EFR	AMR 12,2	AMR 12,2	5	40,8	59,8	88,2	3,08
WB		G.722.2 = AMR-WB	AMR IO 23,85		8	17	131	4,308
SWB			EVS 13,2			17,1	130,9	4,30
FB			EVS 24,4			7,2	140,8	4,44

NOTE 3: As per Recommendation ITU-T G.113 [4]:

- Ie,fb =Ie,wb +19 and
- Ie,wb =Ie,nb +35,8,
- therefore: Ie,fb = Ie,nb + 54,8.

9.6.3 Mean one-way delay MOWD

Recommendation ITU-T P.1010 [14] provides (quality) categories for VoIP terminals in clause 6.2.1 for send delay and in clause 6.2.2 for receive delay.

Summing up the values from Recommendation ITU-T P.1010 [14] results in the following values:

- Category A: MOWD ≤ 50 ms;
- Category B: MOWD ≤ 100 ms;
- Category C: MOWD ≤ 150 ms.

Recommendation ITU-T P.1010 [14] attributes these limitations to the designated categories of VoIP terminals. For the purpose of the present document, 5G UE voice terminals can be considered as VoIP terminals.

For measurement results obtained at local test events without geographical context, any excessive delay should be attributed as unwanted impairment by the network.

NOTE: Of course, any delay exceeding the values stipulated for the categories per Recommendation ITU-T P.1010 [14] may also have its cause in the 5G UE voice terminals. However, a respective root cause analysis is typically out of scope of a test event.

9.6.4 Double-talk

Based on the requirements for send and receive direction of a voice terminal in clause 6.8 of ETSI TS 103 739 [18] the end-to-end connection should exhibit a full duplex capability with an attenuation range during double talk of:

$$A_{H,R,dt} < 6 \text{ dB}.$$

9.6.5 Insertion loss

The nominal value of the insertion loss shall be 0 dB; any deviation thereof shall be considered as an unwanted impairment in the context of 5G test events.

9.6.6 Talker Echo

With reference to Recommendations ITU-T G.131 [6] and G.114 [5] a maximum MOWD of 400 ms can be assumed under regular operational conditions. Therefore, a talker echo resulting from an implemented talker echo loudness rating of TELR > 65 dB should be met under static conditions. Lower values of TELR will have to be considered as an additional impairment.

10 QoS for IoT applications

10.1 Introduction

10.1.1 Objective

The objective of IoT QoS testing in 5G networks is to verify that the end-to-end service behaviour—from the IoT device → RAN → 5G Core → IP Application Server-matches the 5QI-based QoS characteristics such as latency, reliability, priority handling, and packet delivery performance.

Ensuring compliance enables predictable and consistent service behaviour for IoT applications across all network domains.

NOTE: Validation of MQTT delivery behaviour for configured MQTT QoS levels 0,1 and/or 2 are not the scope of testing.

10.1.2 System architecture and QoS parameters (ETSI TS 123 501)

QoS handling and IoT service characteristics are defined at system architecture level:

Table 16

Parameter	Description	3GPP Reference
5QI	QoS characteristics (PDB, PER, priority)	ETSI TS 123 501 [26], clause 5.7
GBR/Non-GBR	Resource type selection	ETSI TS 123 501 [26], clause 5.7
AMBR	Aggregate bitrate limitation	ETSI TS 123 501 [26], clause 5.7
Non-IP Data Delivery (NIDD)	IP-less IoT communication	ETSI TS 123 501 [26], clause 5.31.5
Public Network Integrated NPN	Private IoT network support	ETSI TS 123 501 [26], clause 5.30

10.2 Relevant 5G QoS elements for IoT

5QI - 5G QoS Identifier defines

- Packet Delay Budget (PDB)
- Packet Error Rate (PER)

- Priority Level
- GBR vs. Non-GBR resource type
- RAN and Core scheduling behaviour

ARP - Allocation and Retention Priority

- Determines whether a QoS flow can be admitted or pre-empted during congestion.

AMBR - Aggregated Maximum Bit Rate

- Defines maximum throughput limits for non-GBR IoT traffic.

GBR - Guaranteed Bit Rate

- Used for industrial or critical IoT; provides deterministic minimum throughput.

RAN Scheduling Behaviour

- Enforces prioritization, congestion handling, and differentiation of traffic based on 5QI, ARP, and slice configuration.

10.3 Mapping IoT Use Cases to 5QI and Key KPIs

Table 17

IoT Use Case	Typical 5QI	GBR/Non-GBR	Latency Target	PER	Key KPI Focus	Notes
Periodic sensor data (temperature, humidity, telemetry)	9	Non-GBR	300 ms	10^{-6}	Throughput stability, reliability	Best-effort, low priority
Operational control/actuator commands	7, 8	Non-GBR	100 - 300 ms	10^{-3} - 10^{-6}	Latency, delivery success	Medium priority, moderate reliability
Critical control/industrial IoT	82 - 86	Delay-critical GBR	5 - 30 ms	10^{-4} - 10^{-5}	Low latency, jitter, reliability	High priority, deterministic
Real-time video monitoring IoT	2, 4	GBR	150 - 300 ms	10^{-3} - 10^{-6}	Throughput, latency	High-bandwidth continuous traffic
Alarm/emergency IoT	69, 70, 79	GBR	50 - 200 ms	10^{-5} - 10^{-6}	Guaranteed delivery, latency	Highest-priority flows
NOTE: The 5QI values listed in Table 17 represent typical examples for IoT-oriented deployments. Actual 5QI assignment may depend on operator policy, network slicing configuration, application requirements and PCF-based QoS policies.						

10.4 Essential end-to-end QoS test scenarios

10.4.1 Test Case 1 - latency validation per 5QI

Goal: Ensure measured E2E latency complies with the 5QI Packet Delay Budget (PDB), as specified in Table 18 (from ETSI TS 123 501 [26], table 5.7.4-1).

Method:

- Generate traffic using typical IoT packet sizes.
- Measure one-way and round-trip latency (device ↔ server).
- Compare results with 5QI PDB values.

- Test both idle and congested network states.

Standardized 5QI values are specified for services that are assumed to be frequently used and thus benefit from optimized signalling by using standardized QoS characteristics. Dynamically assigned 5QI values (which require a signalling of QoS characteristics as part of the QoS profile) can be used for services for which standardized 5QI values are not defined. The one-to-one mapping of standardized 5QI values to 5G QoS characteristics is specified in Table 18.

Table 18: Standardized 5QI to QoS characteristics mapping

5QI Value	Resource Type	Default Priority Level	Packet Delay Budget (note 3)	Packet Error Rate	Default Maximum Data Burst Volume (note 2)	Default Averaging Window	Example Services
1	GBR (note 1)	20	100 ms (notes 11 and 13)	10^{-2}	N/A	2 000 ms	Conversational Voice
2		40	150 ms (notes 11 and 13)	10^{-3}	N/A	2 000 ms	Conversational Video (Live Streaming)
3		30	50 ms (notes 11 and 13)	10^{-3}	N/A	2 000 ms	Real Time Gaming, V2X messages (ETSI TS 123 287 [25]). Electricity distribution - medium voltage, Process automation monitoring
4		50	300 ms (notes 11 and 13)	10^{-6}	N/A	2 000 ms	Non-Conversational Video (Buffered Streaming)
65 (notes 9 and 12)		7	75 ms (notes 7 and 8)	10^{-2}	N/A	2 000 ms	Mission Critical user plane Push to Talk voice (e.g. MCPTT)
66 (note 12)		20	100 ms (notes 10 and 13)	10^{-2}	N/A	2 000 ms	Non-Mission-Critical user plane Push to Talk voice
67 (note 12)		15	100 ms (notes 10 and 13)	10^{-3}	N/A	2 000 ms	Mission Critical Video user plane
75 (note 14)							
71		56	150 ms (notes 11, 13 and 15)	10^{-6}	N/A	2 000 ms	"Live" Uplink Streaming (ETSI TS 126 238 [31])
72		56	300 ms (notes 11, 13 and 15)	10^{-4}	N/A	2 000 ms	"Live" Uplink Streaming (ETSI TS 126 238 [31])
73		56	300 ms (notes 11, 13 and 15)	10^{-8}	N/A	2 000 ms	"Live" Uplink Streaming (ETSI TS 126 238 [31])
74		56	500 ms (notes 11 and 15)	10^{-8}	N/A	2 000 ms	"Live" Uplink Streaming (ETSI TS 126 238 [31])
76		56	500 ms (notes 11, 13 and 15)	10^{-4}	N/A	2 000 ms	"Live" Uplink Streaming (ETSI TS 126 238 [31])
5	Non-GBR	10	100 ms (notes 10 and 13)	10^{-6}	N/A	N/A	IMS Signalling
6	(note 1)	60	300 ms (notes 10 and 13)	10^{-6}	N/A	N/A	Video (Buffered Streaming) TCP-based (e.g. www, e-mail, chat, ftp, p2p file sharing, progressive video)
7		70	100 ms (notes 10 and 13)	10^{-3}	N/A	N/A	Voice, Video (Live Streaming) Interactive Gaming

5QI Value	Resource Type	Default Priority Level	Packet Delay Budget (note 3)	Packet Error Rate	Default Maximum Data Burst Volume (note 2)	Default Averaging Window	Example Services
8		80	300 ms (note 13)	10^{-6}	N/A	N/A	Video (Buffered Streaming) TCP-based (e.g. www, e-mail, chat, ftp, p2p file sharing, progressive video)
9		90					
10		90	1 100 ms (notes 13 and 17)	10^{-6}	N/A	N/A	Video (Buffered Streaming) TCP-based (e.g. www, e-mail, chat, ftp, p2p file sharing, progressive video, etc.) and any service that can be used over satellite access type with these characteristics
69 (notes 9 and 12)		5	60 ms (notes 7 and 8)	10^{-6}	N/A	N/A	Mission Critical delay sensitive signalling (e.g. MCPTT signalling)
70 (note 12)		55	200 ms (notes 7 and 10)	10^{-6}	N/A	N/A	Mission Critical Data (e.g. example services are the same as 5QI 6/8/9)
79		65	50 ms (notes 10 and 13)	10^{-2}	N/A	N/A	V2X messages (ETSI TS 123 287 [25])
80		68	10 ms (notes 5 and 10)	10^{-6}	N/A	N/A	Low Latency eMBB applications Augmented Reality
82	Delay-critical GBR	19	10 ms (note 4)	10^{-4}	255 bytes	2 000 ms	Discrete Automation (ETSI TS 122 261 [21])
83		22	10 ms (note 4)	10^{-4}	1 354 bytes (note 3)	2 000 ms	Discrete Automation (ETSI TS 122 261 [21]); V2X messages (UE - RSU Platooning, Advanced Driving: Cooperative Lane Change with low Load. ETSI TS 122 186 [20], ETSI TS 123 287 [25])
84		24	30 ms (note 6)	10^{-5}	1 354 bytes (note 3)	2 000 ms	Intelligent transport systems (ETSI TS 122 261 [21])
85		21	5 ms (note 5)	10^{-5}	255 bytes	2 000 ms	Electricity Distribution - high voltage (ETSI TS 122 261 [21]); V2X messages (Remote Driving. ETSI TS 122 186 [20], note 16, ETSI TS 123 287 [25])

5QI Value	Resource Type	Default Priority Level	Packet Delay Budget (note 3)	Packet Error Rate	Default Maximum Data Burst Volume (note 2)	Default Averaging Window	Example Services
86		18	5 ms (note 5)	10^{-4}	1 354 bytes	2 000 ms	V2X messages (Advanced Driving: Collision Avoidance, Platooning with high Loa. ETSI TS 122 186 [20], ETSI TS 123 287 [25])
87		25	5 ms (note 4)	10^{-3}	500 bytes	2 000 ms	Interactive Service - Motion tracking data, (ETSI TS 122 261 [21])
88		25	10 ms (note 4)	10^{-3}	1 125 bytes	2 000 ms	Interactive Service - Motion tracking data, (ETSI TS 122 261 [21])
89		25	15 ms (note 4)	10^{-4}	17 000 bytes	2 000 ms	Visual content for cloud/edge/split rendering (ETSI TS 122 261 [21])
90		25	20 ms (note 4)	10^{-4}	63 000 bytes	2 000 ms	Visual content for cloud/edge/split rendering (ETSI TS 122 261 [21])

5QI Value	Resource Type	Default Priority Level	Packet Delay Budget (note 3)	Packet Error Rate	Default Maximum Data Burst Volume (note 2)	Default Averaging Window	Example Services

NOTE 1: A packet which is delayed more than PDB is not counted as lost, thus not included in the PER.
NOTE 2: It is required that default MDBV is supported by a PLMN supporting the related 5QIs.
NOTE 3: The Maximum Transfer Unit (MTU) size considerations in clause 9.3 and Annex C of ETSI TS 123 060 [22] are also applicable. IP fragmentation may have impacts to CN PDB.
NOTE 4: A static value for the CN PDB of 1 ms for the delay between a UPF terminating N6 and a 5G-AN should be subtracted from a given PDB to derive the packet delay budget that applies to the radio interface.
NOTE 5: A static value for the CN PDB of 2 ms for the delay between a UPF terminating N6 and a 5G-AN should be subtracted from a given PDB to derive the packet delay budget that applies to the radio interface.
NOTE 6: A static value for the CN PDB of 5 ms for the delay between a UPF terminating N6 and a 5G-AN should be subtracted from a given PDB to derive the packet delay budget that applies to the radio interface.
NOTE 7: For Mission Critical services, it may be assumed that the UPF terminating N6 is located "close" to the 5G-AN (roughly 10 ms) and is not normally used in a long distance, home routed roaming situation. Hence a static value for the CN PDB of 10 ms for the delay between a UPF terminating N6 and a 5G-AN should be subtracted from this PDB to derive the packet delay budget that applies to the radio interface.
NOTE 8: In both RRC Idle and RRC Connected mode, the PDB requirement for these 5QIs can be relaxed (but not to a value greater than 320 ms) for the first packet(s) in a downlink data or signalling burst in order to permit reasonable battery saving (DRX) techniques.
NOTE 9: It is expected that 5QI-65 and 5QI-69 are used together to provide Mission Critical Push to Talk service (e.g. 5QI-5 is not used for signalling). It is expected that the amount of traffic per UE will be similar or less compared to the IMS signalling.
NOTE 10: In both RRC Idle and RRC Connected mode, the PDB requirement for these 5QIs can be relaxed for the first packet(s) in a downlink data or signalling burst in order to permit battery saving (DRX) techniques.
NOTE 11: In RRC Idle mode, the PDB requirement for these 5QIs can be relaxed for the first packet(s) in a downlink data or signalling burst in order to permit battery saving (DRX) techniques.
NOTE 12: This 5QI value can only be assigned upon request from the network side. The UE and any application running on the UE is not allowed to request this 5QI value.
NOTE 13: A static value for the CN PDB of 20 ms for the delay between a UPF terminating N6 and a 5G-AN should be subtracted from a given PDB to derive the packet delay budget that applies to the radio interface.
NOTE 14: This 5QI is not supported as it is only used for transmission of V2X messages over MBMS bearers as defined in ETSI TS 123 285 [24] but the value is reserved for future use.
NOTE 15: For "live" uplink streaming (ETSI TS 126 238 [31]), guidelines for PDB values of the different 5QIs correspond to the latency configurations defined in ETSI TR 126 939 [i.3]. In order to support higher latency reliable streaming services (above 500ms PDB), if different PDB and PER combinations are needed these configurations will have to use non-standardized 5QIs.
NOTE 16: These services are expected to need much larger MDBV values to be signalled to the RAN.
NOTE 17: The worst case one way propagation delay for GEO satellite is expected to be ~270 ms,~ 21 ms for LEO at 1 200 km, and 13 ms for LEO at 600 km. The UL scheduling delay that needs to be added is also typically two way propagation delay e.g. ~540 ms for GEO, ~42 ms for LEO at 1 200 km, and ~26 ms for LEO at 600 km. Based on that, the 5G-AN Packet delay budget is not applicable for 5QIs that require 5G-AN PDB lower than the sum of these values when the specific types of satellite access are used (see ETSI TS 138 300 [32]). 5QI-10 can accommodate the worst case PDB for GEO satellite type.

10.4.2 Test Case 2 - Reliability & Packet Error Rate (PER)

Goal: Verify that observed PER aligns with 5QI target values.

Method:

- Perform repeated uplink and downlink message transmission.
- Measure packet loss and retransmissions.
- Validate performance under mild congestion conditions.

10.4.3 Test Case 3 - Congestion & Priority handling

Goal: Validate correct prioritization and scheduling across different 5QIs.

Method:

- Introduce controlled RAN/Core congestion.
- Run parallel flows with different 5QI values.
- Measure latency, jitter, and packet loss under load.
- Validate ARP behaviour and confirm that high-priority flows are not pre-empted.

10.4.4 Test Case 4 - Throughput & AMBR/GBR enforcement

Goal: Ensure bandwidth limits and guaranteed bit rates are correctly enforced.

Method:

- Apply sustained traffic load.
- Observe throughput for GBR and non-GBR flows.
- Verify that non-GBR flows respect AMBR limits.
- Ensure GBR flows maintain guaranteed throughput during congestion.

10.5 Material to be prepared for test events

Unified **IoT Use Case → 5QI → KPI → Test Case** mapping.

Traffic generation templates for each IoT service type.

Automated scripts for:

- Latency measurement
- Packet error/loss analysis
- Congestion simulation

Result sheets aligned with:

- ETSI TS 123 501 [26]
- ETSI TS 123 503 [28]
- ETSI TS 124 501 [30]

Summary scoring and compliance analysis for E2E QoS performance.

10.6 Application specific QoS adaptors

For the measurement of parameter of IoT applications, specific adaptors will be needed to access the relevant functional trigger points.

Since these trigger points are not commonly standardized, adaptors for QoS relevant trigger points will either have to be provided by application provider or by the test house.

NOTE: Adaptors may also jointly be defined by the IoT application provider and the test house. In case, care has to be taken to access the proper trigger pointing a non-intrusive manner.

10.7 Reporting

The test house shall prepare a detailed test report, including details of:

- the IoT application under test;
- condition(s) of the network;
- details of the trigger points;
- adaptor properties;
- statistical analysis of the results;
- educated guess about quality on comparison scale;
- etc.

Materials prepared before the test event according to clause 10.5 and completed by the results obtained shall be included as electronic attachment to the test report.

11 Test Descriptions

11.1 VoNR Test Descriptions with QoS check

11.1.1 Call establishment over gNB

Interoperability Test Description			
Identifier:	TD_VoNR_CALL_01		
Objective:	Demonstrate the call establishment between originating UE and terminating UE and perform QoS voice measurement.		
Configuration:	CFG_VoNR_INT_1HNW CFG_VoNR_INT_2HNW CFG_VoNR_RMI_2NW_A CFG_VoNR_RMI_2NW_B		
References:	SIP - ETSI TS 124 229 [29] NGAP/NAS - ETSI TS 123 502 [27]		
Pre-test conditions:	• Originating and Terminating UE registered to 5GC (see clause 8.1.1) over gNB. • 5GC established a Default PDU session (see clause 8.1.2.2) allowing UE to P-CSCF IP communication. • Originating and Terminating UE registered to IMS (see clause 8.1.2.2). • 5GC established an IMS signalling PDU session. • Terminating UE ready to accept the session establishment.		
Test Sequence:	**Step**	**Type**	**Description**
	1	verify	Media between originating UE and terminating UE is not delivered in any direction before call establishment.
	2	check	Originating UE calls terminating UE and establishes a communication session (see clause 8.1.4).
	3	verify	Verify that media between originating UE and terminating UE is successfully routed over the dedicated PDU session.
	4	obtain result	QoS test clause 9.4.2 MOS-LQOF (mandatory).
	5	verify advanced	If advanced test environment (Annex B) is available proceed with step 6 else: skip to step 10.
	6	obtain result	QoS test clause 9.4.3 Delay.
	7	obtain result	QoS test clause 9.4.4 Double Talk.
	8	obtain result	QoS test clause 9.4.5 Insertion Loss.
	9	obtain result	QoS test clause 9.4.6 Talker Echo.
	10	create	Statistical analysis and reporting (clause 9.5).
	11	conduct	Classification of results (clause 9.6).

11.1.2 Handover after call establishment

Interoperability Test Description			
Identifier:	TD_VoNR_CALL_02		
Objective:	Demonstrate the call establishment between originating UE and terminating UE. Change location of originating UE during the call to a target gNB and perform QoS voice measurement.		
Configuration:	CFG_VoNR_INT_HO_1HNW		
References:	SIP - ETSI TS 124 229 [29] NGAP/NAS - ETSI TS 123 502 [27]		
Pre-test conditions:	• Originating and Terminating UE registered to 5GC (see clause 8.1.1) over gNB. • 5GC established a Default PDU session (see clause 8.1.2.2) allowing UE to P-CSCF IP communication. • Originating and Terminating UE registered to IMS (see clause 8.1.2.2). • 5GC established an IMS signalling PDU session. • Terminating UE ready to accept the session establishment.		
Test Sequence:	**Step**	**Type**	**Description**
	1	verify	Media between originating UE and terminating UE is not delivered in any direction before call establishment.
	2	check	Originating UE calls terminating UE and establishes a communication session (see clause 8.1.4).
	3	verify	Verify that media between originating UE and terminating UE is successfully routed over the dedicated PDU session.
	4	check	Originating UE changes its location and a handover to a target gNB is successfully performed while maintaining the communication session.
	5	verify	Verify that media between originating UE and terminating UE is still active over the dedicated PDU session.
	6	obtain result	QoS test clause 9.4.2 MOS-LQOF (mandatory).
	7	verify advanced	If advanced test environment (Annex B) is available proceed with step 8 else: skip to step 12.
	8	obtain result	QoS test clause 9.4.3 Delay.
	9	obtain result	QoS test clause 9.4.4 Double Talk.
	10	obtain result	QoS test clause 9.4.5 Insertion Loss.
	11	obtain result	QoS test clause 9.4.6 Talker Echo.
	12	create	Statistical analysis and reporting (clause 9.5).
	13	conduct	Classification of results (clause 9.6).

11.1.3 UE switched from gNB to WLAN after call establishment

Interoperability Test Description			
Identifier:	TD_VoNR_CALL_03		
Objective:	Demonstrate the call establishment between originating UE and terminating UE. Originating UE switch on WLAN connectivity during the call and perform QoS voice measurement.		
Configuration:	CFG_VoNR_INT_WLAN_1HNW CFG_VoNR_INT_2HNW CFG_VoNR_RMI_2NW_A CFG_VoNR_RMI_2NW_B		
References:	SIP - ETSI TS 124 229 [29] NGAP/NAS - ETSI TS 123 502 [27]		
Pre-test conditions:	• Originating and Terminating UE registered to 5GC (see clause 8.1.1) over gNB. • 5GC established a Default PDU session (see clause 8.1.2.2) allowing UE to P-CSCF IP communication. • Originating and Terminating UE registered to IMS (see clause 8.1.2.2). • 5GC established an IMS signalling PDU session. • Terminating UE ready to accept the session establishment.		
Test Sequence:	**Step**	**Type**	**Description**
	1	verify	Media between originating UE and terminating UE is not delivered in any direction before call establishment.
	2	check	Originating UE calls terminating UE and establishes a communication session (see clause 8.1.4).
	3	verify	Verify that media between originating UE and terminating UE is successfully routed over the dedicated PDU session.
	4	check	Originating UE switch on WLAN connectivity during the call while maintaining the communication session.
	5	verify	Verify that media between originating UE and terminating UE is still active over the dedicated PDU session.
	6	obtain result	QoS test clause 9.4.2 MOS-LQOF (mandatory).
	7	verify advanced	If advanced test environment (Annex B) is available proceed with step 8 else: skip to step 12.
	8	obtain result	QoS test clause 9.4.3 Delay.
	9	obtain result	QoS test clause 9.4.4 Double Talk.
	10	obtain result	QoS test clause 9.4.5 Insertion Loss.
	11	obtain result	QoS test clause 9.4.6 Talker Echo.
	12	create	Statistical analysis and reporting (clause 9.5).
	13	conduct	Classification of results (clause 9.6).

11.2 IoT Test Descriptions

11.2.1 MQTT communication over 5GC

Interoperability Test Description			
Identifier:	TD_IoT_5GEST_01		
Objective:	Demonstrate end-to-end IoT communication between IoT device(s) and application server using MQTT protocol over 5G system and perform QoS validation (latency, reliability, throughput).		
Configuration:	CFG_5GSA CFG_5GSA_SNPN_PLMN (optional) CFG_5GSA_PLMN_SNPN (optional)		
References:	MQTT - OASIS MQTT Version 5.0 [35] 5G NAS/NGAP - ETSI TS 123 502 [27] 5G QoS - ETSI TS 123 501 [26]		
Pre-test conditions:	• IoT device(s) registered to 5GC (see clause 8.2.1). • PDU session established (see clause 8.2.2) to allow connection towards Data Network (DN) supporting IoT application server. • MQTT broker reachable through the Data Network (DN) via the UPF over the N6 interface. • Application server connected to the MQTT broker and subscribed to the relevant topics. • IoT device configured for MQTT publish/subscribe communication. • QoS profile (5QI, AMBR, GBR where applicable) configured in 5GC. • Application server subscribed to relevant MQTT topics.		
Test Sequence:	**Step**	**Type**	**Description**
	1	verify	No MQTT data exchange occurs before session establishment.
	2	check	IoT device establishes connection to MQTT broker (CONNECT/CONNACK).
	3	check	IoT device publishes MQTT message(s) to configured topic(s).
	4	verify	Verify that MQTT messages are successfully delivered to application server (PUB/SUB flow).
	5	verify	Verify correct routing of traffic over established PDU session and QoS flow (based on 5QI).
	6	obtain result	QoS test: End-to-end latency measurement (clause 10.4.1).
	7	obtain result	QoS test: Packet Error Rate/reliability (clause 10.4.2).
	8	obtain result	QoS test: Throughput and AMBR/GBR enforcement (clause 10.4.4).
	9	check	Introduce congestion and parallel flows with different priorities.
	10	verify	Verify correct prioritization and QoS handling under congestion conditions (clause 10.4.3).
	11	verify advanced	If advanced test environment available: • proceed with step 12; • else: skip to step 14.
	12	obtain result	QoS test: latency under congestion conditions.
	13	obtain result	QoS test: message delivery success under load.
	14	create	Statistical analysis and reporting (clause 10.7).
	15	conduct	Classification of results against 5QI KPIs (clauses 10.3 and 10.4).

11.2.2 CoAP communication over 5GC

Interoperability Test Description			
Identifier:	TD_IoT_5GEST_02		
Objective:	Demonstrate end-to-end IoT communication between IoT device(s) and application server using Constrained Application Protocol (CoAP) over the 5G system and perform QoS validation for latency, reliability and packet delivery.		
Configuration:	CFG_5GSA CFG_5GSA_SNPN_PLMN (optional) CFG_5GSA_PLMN_SNPN (optional)		
References:	CoAP - IETF RFC 7252 [36] 5G NAS/NGAP - ETSI TS 123 502 [27] 5G QoS - ETSI TS 123 501 [26]		
Pre-test conditions:	• IoT device(s) registered to 5GC (see clause 8.2.1). • PDU session established (see clause 8.2.2) to allow connection towards Data Network (DN) supporting IoT application server. • CoAP server reachable via N6 interface. • IoT device configured for CoAP client/server communication. • QoS profile (5QI, AMBR, GBR where applicable) configured in 5GC. • Application server configured to receive CoAP requests and responses.		
Test Sequence:	**Step**	**Type**	**Description**
	1	verify	No CoAP data exchange occurs before session establishment.
	2	check	IoT device establishes communication with CoAP server.
	3	check	IoT device transmits CoAP Confirmable (CON) and/or Non-confirmable (NON) messages.
	4	verify	Verify that CoAP messages are successfully delivered between IoT device and application server.
	5	verify	Verify correct routing of traffic over established PDU session and QoS flow based on configured 5QI.
	6	obtain result	QoS test: End-to-end latency measurement (mandatory) according to clause 10.4.1.
	7	obtain result	QoS test: Packet Error Rate and retransmission behaviour according to clause 10.4.2.
	8	obtain result	QoS test: Throughput and AMBR/GBR enforcement according to clause 10.4.4.
	9	check	Introduce congestion and parallel flows with different priorities.
	10	verify	Verify correct prioritization and QoS handling under congestion conditions (clause 10.4.3).
	11	verify advanced	If advanced test environment is available proceed with step 12 else: skip to step 14.
	12	obtain result	QoS test: latency under congestion conditions.
	13	obtain result	QoS test: message delivery success under load.
	14	create	Statistical analysis and reporting (clause 10.7).
	15	conduct	Classification of results against 5QI KPIs (clauses 10.3 and 10.4).

11.2.3 HTTP/REST communication over 5GC

Interoperability Test Description			
Identifier:	TD_IoT_5GEST_03		
Objective:	Demonstrate end-to-end IoT communication between IoT device(s) and application server using HTTP/REST services over the 5G system and perform QoS validation for latency, reliability and throughput.		
Configuration:	CFG_5GSA CFG_5GSA_SNPN_PLMN (optional) CFG_5GSA_PLMN_SNPN (optional)		
References:	HTTP - IETF RFC 9110 [37] 5G NAS/NGAP - ETSI TS 123 502 [27] 5G QoS - ETSI TS 123 501 [26]		
Pre-test conditions:	• IoT device(s) registered to 5GC (see clause 8.2.1). • PDU session established (see clause 8.2.2) to allow connection towards Data Network (DN) supporting IoT application server. • HTTP/REST application server reachable via N6 interface. • IoT device configured for HTTP/REST communication. • QoS profile (5QI, AMBR, GBR where applicable) configured in 5GC. • Application server configured to process HTTP GET/POST/PUT requests.		
Test Sequence:	**Step**	**Type**	**Description**
	1	verify	No HTTP/REST data exchange occurs before session establishment.
	2	check	IoT device establishes communication with HTTP/REST application server.
	3	check	IoT device exchanges HTTP/REST messages with the application server.
	4	verify	Verify successful exchange of application payload data between IoT device and application server.
	5	verify	Verify correct routing of traffic over established PDU session and QoS flow based on configured 5QI.
	6	obtain result	QoS test: End-to-end latency measurement (mandatory) according to clause 10.4.1.
	7	obtain result	QoS test: Packet Error Rate and reliability according to clause 10.4.2.
	8	obtain result	QoS test: Throughput and AMBR/GBR enforcement according to clause 10.4.4.
	9	check	Introduce congestion and parallel flows with different priorities.
	10	verify	Verify correct prioritization and QoS handling under congestion conditions (clause 10.4.3).
	11	verify advanced	If advanced test environment is available proceed with step 12 else: skip to step 14.
	12	obtain result	QoS test: latency under congestion conditions.
	13	obtain result	QoS test: application response success under load.
	14	create	Statistical analysis and reporting (clause 10.7).
	15	conduct	Classification of results against 5QI KPIs (clauses 10.3 and 10.4).

Annex A (informative): Template for co-ordination of QoS options

This annex provides in Table A.1 an example template for the various QoS and QoE related options with application columns per stakeholder.

Table A.1: Example template

Topic	Option	Vendor/Manufacturer	Test event organizer	Test house	Remarks
UE voice bandwidth	Narrowband				
	Wideband				
	Super-Wideband				
	Fullband			X (mandatory)	
UE voice enhancement	Acoustic Echo Cancellation				
	Adaptive Noise Suppression				
Voice Terminal, user interface	Acoustical				
	Bluetooth®				
	Headset electrical				
	Electrical other				
	Optical				
Voice Terminal, network interface	5G VoNR	X			
	VoWLAN	X			
Network	Simulation of packet loss				
	Simulation of jitter				
	Background load				
	Traffic generation				
QoS/QoE parameter	End-to-End Listening Quality MOS-LQOF	X	X	X	X (mandatory)
	Mean one-way Delay				
	Double Talk capability				
	Insertion Loss				
	Talker Echo				

Annex B (informative): Advanced environmental test conditions

The straightforward assessment of listening quality MOS-LQOF can be done in an offline mode if preferred. During the test event speech files are inserted into the send side UE and the distorted speech files are obtained from the receive side UE. Processing can be done offline after the event.

However, already the assessment of mean one-way delay MOWD requires a real-time recording and analysis.

The assessment of the other parameter mentioned in clause 9.6:

- double talk;
- insertion loss;
- talker echo

requires even more advanced environmental test conditions as e.g. outlined in clause 5.3 of ETSI TS 103 739 [18], which stipulates that in general different acoustical environments have to be taken into account: either room noise and background noise are an inherent part of the test environment or room noise and background noise should be eliminated to such an extent that their influence on the test results can be neglected.

Furthermore and unless stated otherwise, measurements should be conducted under quiet and "anechoic" conditions. Considering this, test laboratory, in the case where its test room does not conform to anechoic conditions as given in Recommendation ITU-T P.310 [8], has to present difference in results for measurements due to its test room. In case where an anechoic room is not available the test room has to be an acoustically treated room with few reflections and a low noise level.

Depending on the distance of the transducers from mouth to ear a quiet office room may be sufficient e.g. for handsets where artificial mouth and artificial ear are located close to the acoustical transducers.

However, for some headsets or handset terminals with smaller dimension an anechoic room will be required.

In cases where real or simulated background noise is used as part of the testing environment, the original background noise should not be noticeably influenced by the acoustical properties of the room.

In all cases where the performance of acoustic echo cancellers should be tested a realistic room which represents the typical user environment for the terminal should be used.

NOTE 1: This means in other words that also rooms which do not fulfil the formal requirements to 100 % may be used under the condition that the deviation parameters have been tested and documented.

NOTE 2: For the purpose of test events which involve both network interoperability aspects and QoS/QoE (as in the present document), the use of portable anechoic chambers may be suitable.

Annex C (informative): Bibliography

C.1 General QoS/QoE aspects

- Quality of service regulation manual, ITU Geneva, 2017, ISBN 978-92-61-25791-0 (electronic version).
- Quality of Service Training Programme, ITU Academy, Geneva, December 2014.
- Recommendation ITU-T G.1011 (07/2016): "Reference guide to quality of experience assessment methodologies".
- Recommendation ITU-T P.800.1 (07/2016): "Mean opinion score (MOS) terminology".

C.2 Objective prediction of listening quality

- J. G. Beerends, C. Schmidmer, J. Berger, M. Obermann, R. Ullmann, J. Pomy, and M. Keyhl: "Perceptual Objective Listening Quality Assessment (POLQA), The Third Generation ITU-T Standard for End-to-End Speech Quality Measurement, Part I—Temporal Alignment", Journal of the Audio Engineering Society, vol. 61, no. 6, pp. 366-384, 2013.
- J. G. Beerends, C. Schmidmer, J. Berger, M. Obermann, R. Ullmann, J. Pomy, and M. Keyhl: "Perceptual Objective Listening Quality Assessment (POLQA), The Third Generation ITU-T Standard for End-to-End Speech Quality Measurement, Part II—Perceptual Model", Journal of the Audio Engineering Society, vol. 61, no. 6, pp. 385-402, 2013.
- POLQA, Technical White Paper, www.polqa.info

C.3 E-Model

- E-Model tutorial, ITU-T Geneva.
- Updated E-Model online calculation tool: http://www.e-model.org/emodel-2019-11/index.htm

C.4 VoLTE

- Recommendation ITU-T G.1028: "End-to-end quality of service for voice over 4G mobile networks".
- Recommendation ITU-T G.1028.2: "Assessment of the LTE circuit switched fall back - Impact on voice quality of service".

History

Version	Date	Status
V1.1.1	August 2026	Publication